

Multycomm

User Manual



Version 3.1

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Introduction

Multycomm Unified Communications Service Platform, is a scalable, managed, turnkey solution for carriers and service providers, designed to provide multi-tenant business communications, next generation office PBX, including voice, video messaging and presence to Enterprises, SMB and Residential markets.

- Fully Redundant at each component level
 - Each component of the system can be scaled across different servers and geographically separated data centers.
- True Multi-Tenant
 - Built from ground up as a multi-tenant horizontally scalable system.
 - Does not rely on virtualization to achieve multi-tenancy.
- Turn-Key and Managed
 - Customized installation, configuration and training.
 - Includes NOC monitoring, alerting and dashboards

Multycomm Features

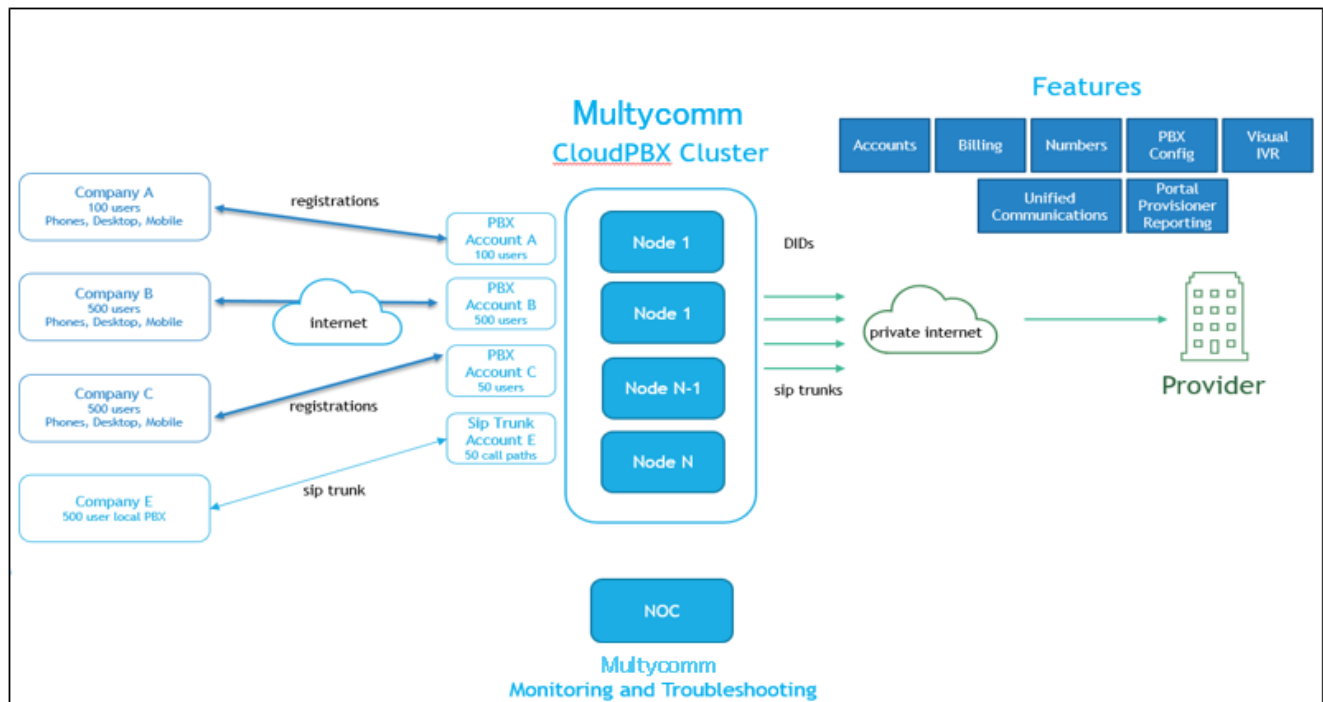
Account Management	Dial-by-Name Directory	Number Porting
Directory Service	Do Not Disturb	Queues
Bulk Number Porting	E911 Dialing	Ring-Ready Phones
Call Forwarding	Fax From Phone	Schedules
Find Me Rules	Follow Me	Text Messaging (SMS)
Call Notification	Greetings	Toll Free Phone Numbers
Call Screening	HD Voice	Canadian Based Support
Call Transfer	Hold Music	User Extensions
Caller ID (Incoming)	In-Call Features	Voicemail
Caller ID (Outgoing)	Intuitive Web Interface	Voicemail to Email
Soft Phone with Zoiper	Local & Long Distance	Email to Fax
Custom Local Numbers	Local Numbers	User Control Panel
Custom Toll Free Number	Menus (IVR)	Routing / Hunt Groups

Theory of Operation

Multycmm platform is provisioned/installed as a cluster across a minimum of three servers in a customer's private cloud. Each server has identical configuration, thus there is no single point of failure.

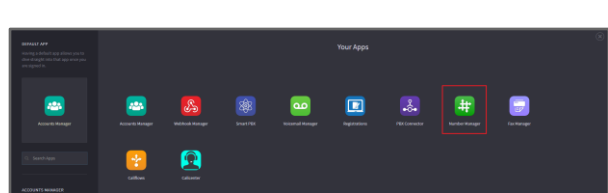
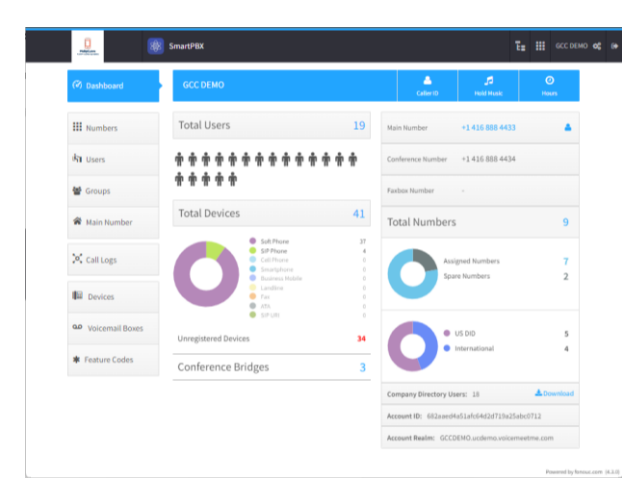
Multycm Core Components

- SIP Router/Registrar
- Media Server
- Call Control Applications
- Distributed Clustered Database
- High capacity message broker
- Web Server
- Security and Monitoring
- Multycmm Web GUI



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- System Admin Portal: <https://ucdemo.voicemeetme.com/portal/login>
 - Example: <https://ucdemo.voicemeetme.com/portal>
 - System administration
 - Account creation
 - Number importing
 - User and Admin Onboarding
 - Phone provisioning
 - Reports and Analytics
- SmartPBX Portal: <https://ucdemo.voicemeetme.com/>
 - Example: <https://ucdemo.voicemeetme.com>
 - Easy to use, self-service tenant PBX administration
 - SmartPBX
 - Quick PBX configurator
 - Callflow:
 - Advanced PBX configuration including visual IVR builder
- NOC Portal for Admins
 - Monitoring, Alerting, Reporting



Account

Tenant Accounts

LDAP Server

Directories

Wholesale

Tags

Softphones

Security

Phone System

Storage

Numbers

Recordings

Billing

Integrations

CRM

Accounts

Filters

Name

Filter by Account Status

Working Account

Credit Balance

2529.60

Threshold Alert

900.00

Recharge Threshold

1000.00

Recharge Amount

1000.00

Name	Domain	Users	Devices	Status	Credit	Call Limits	Service Plan	Features	Actions
Anant_customer_demo	Anant_customer_demo.voicemeet.com	4	4	Active	100.00	1	service_plan_default	</	



Multi-Tenant Accounts

Multycomm is a multi-tenant system that supports multiple nested account hierarchies. Each account is an independent customer with its own unique realm/domain. An account can be configured as a pbx user or a reseller. A reseller is a special account that can create its own customers/accounts.

- Master Account
 - Customer 1
 - Customer 2
 - Customer 3
 - Reseller 1
 - Reseller 1's Customer 1
 - Reseller 1's Customer 2
 - Reseller 1's Reseller A
 - Reseller A's Customer 1
 - Reseller A's Customer 2
 - Reseller 2
 - Reseller 2's Customer 1

- Reseller 2's Customer 2

Master Admin Account

This is the top level account and is the first created account on the Multycomm system. This account has special privileges and is able to create subsequent accounts. The service provider would use this account to create new or manage existing accounts for their customers.

- Full admin/root privileges
- Account Management
 - Create new accounts
 - Make an account a reseller
 - Modify or Delete accounts
 - Login as (masquerade) as other accounts
 - Add and remove funds from an account
- Phone Number Management
 - Add numbers/DIDs into the Multycomm platform via GUI Number Manager
Note: only Master Admin account users can add DIDs into the Multycomm platform.
- Carrier Management
 - Define peering/trunking gateways to external sip trunk and DID/phone number providers.

Standard Account

Standard Account is considered an “end-user” account owned by the end customers. Each account has two types of users: Admin and Normal users. The customer would self provision their PBX using the Admin user privileges on the Standard Account. Alternatively the service provider's onboarding team would use the Master Admin Account: Admin user to log into the customer's account as an Admin and provision the customer's PBX.

Reseller Account

Reseller account is a special “Standard Account” that has been elevated to a Reseller level by the Master Admin. The Reseller “owns” his/her customers, thus Reseller is able to create Standard “sub” accounts under the Reseller account for each of his/her end customers.

Account

Tenant Accounts

LDAP Server

Directories

Whitelabel

Tags

Softphones

Add Account

Set Up

General Account

Account Name *

Resalm *

Is Reseller

Timezone *

☐

America/Los_Angeles

Create a new Cloud PBX Account

In this section we will outline all the steps necessary to create a new customer/tenant account and provision cloud PBX services for a new customer.¹

Name	Domain	Users	Devices	Status	Credit	Call Limits	Service Plan	Features	Actions
Anant_customer_demo	Anant_customer_demo.ucdemo.voicemeetme.com	4	4	Active	100.00	1	service_plan_default	[Icons]	[Edit] [Delete]
Carneval_demo	Carneval_demo.ucdemo.voicemeetme.com	7	2	Active	100.00	2	service_plan_default	[Icons]	[Edit] [Delete]
demo Cst	demo.ucdemo.voicemeetme.com	12	24	Active	100.00	8	service_plan_default	[Icons]	[Edit] [Delete]
EXSTAF_DEMO_FILAMENT	EXSTAF_DEMO_FILAMENT.ucdemo.voicemeetme.com	3	6	Active	100.00	2	service_plan_default	[Icons]	[Edit] [Delete]
fonosUC	fonosUC.ucdemo.voicemeetme.com	3	6	Active	100.00	2	service_plan_default	[Icons]	[Edit] [Delete]
GCC DEMO	GCC DEMO.ucdemo.voicemeetme.com	19	37	Active	99.40	2	service_plan_default	[Icons]	[Edit] [Delete]
GlobeHunters_Demo	GlobeHunters_Demo.ucdemo.voicemeetme.com	12	22	Active	991.00	4	service_plan_default	[Icons]	[Edit] [Delete]
HUUVIETNAM DEMO	HUUVIETNAM DEMO.ucdemo.voicemeetme.com	3	0	Active	100.00	2	service_plan_default	[Icons]	[Edit] [Delete]
HYPERCOM DEMO	HYPERCOM DEMO.ucdemo.voicemeetme.com	4	6	Active	100.00	2	service_plan_default	[Icons]	[Edit] [Delete]
ITCOM DEMO	ITCOM DEMO.ucdemo.voicemeetme.com	5	6	Active	100.00	0	service_plan_default	[Icons]	[Edit] [Delete]

Summary

1. Log into the Admin Portal
 - a. Example: <https://ucdemo.voicemeetme.com/portal/configuration>
 - b. Using Master or Reseller Admin Account
 - c. Note the top account selector
2. Select Config -> Accounts
3. Click on
4. Specify account Name and unique Realm
 - a. **Note:** It is suggested that Realm name be related to the account name for easy of future provisioning and debugging
5. Specify the account address and administrator
6. Configure the Service Plan
 - a. Choose to overwrite service plan options, such as custom pricing.
7. Configure PBX Limits
 - a. How many two way simultaneous calls
 - b. Disable post-pay, and enable pre-pay as part of fraud prevention on metered calling.
8. Configure Fraud limits based on maximum daily rate usage

¹ This section assumes that the Multyscomm team has already installed and preconfigured the Multyscomm cluster with Service Plans, Rate Decks, Carrier Gateways and confirmed that all sections of the Multyscomm system are operational and functioning correctly.

Account

Tenant Accounts

LDAP Server

Directories

Whitelabel

Tags

Selfphones

Security

Phone System

Storage

Numbers

Recordings

Bitting

Integrations

CRM

Reports

Provision

Status

Config

Users

SmartPilot

Multi User Session

admin

Add Account

Set Up

General Account

Account Name

Region

Is Reseller

Timezone

Contact Information

Technical

Email

Name

Number

Company Address

Country

Region

Locality

Street Address

ZIP Code

Service Plan

Service Plan

(+) Override Service Plan

Collaboration

State

Enabled

Auto Provisioning

UCP

Enabled

Select Audio Codes

UCP Mobile

Enabled

Select Audio Codes

Select Video Codes

Call Recording

State

Enabled

Call Restrictions

Limits

Two-way trunks

Inbound trunks

Outbound trunks

Config

Retention Duration

Credit

Add

Threshold Alert

Recharge Threshold

Recharge Amount

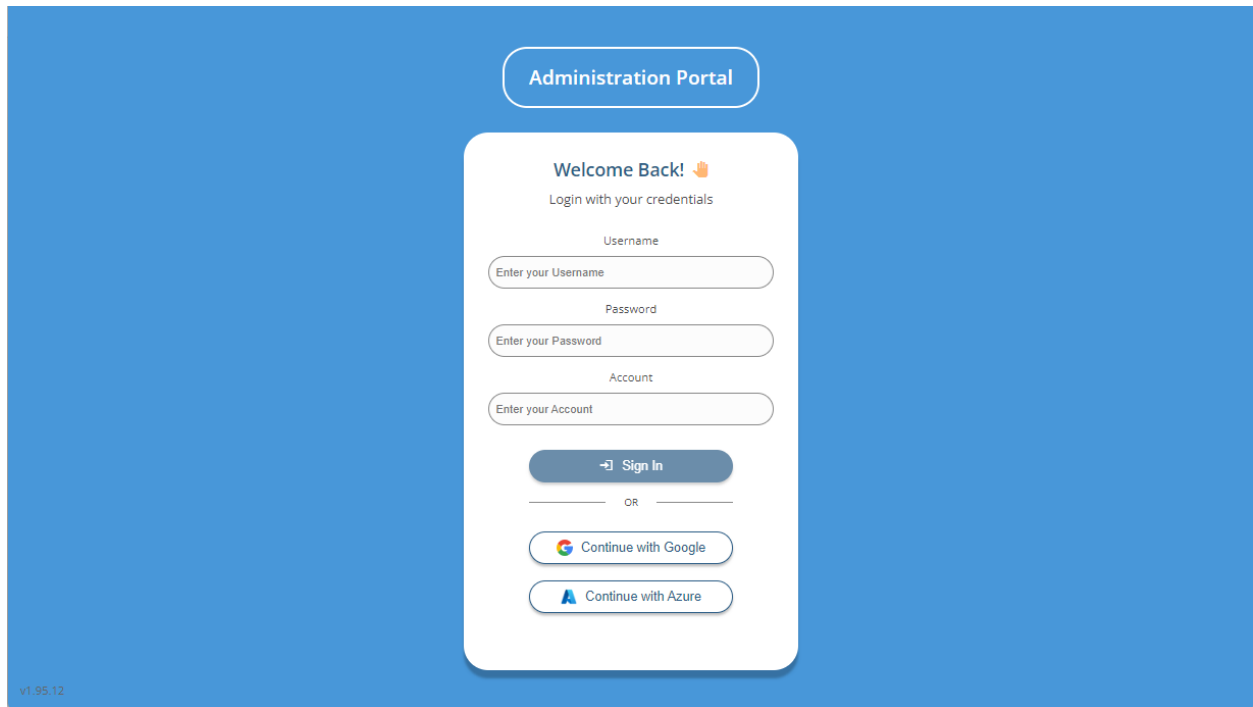
Max Daily Fraud Risk

Cancel

Submit

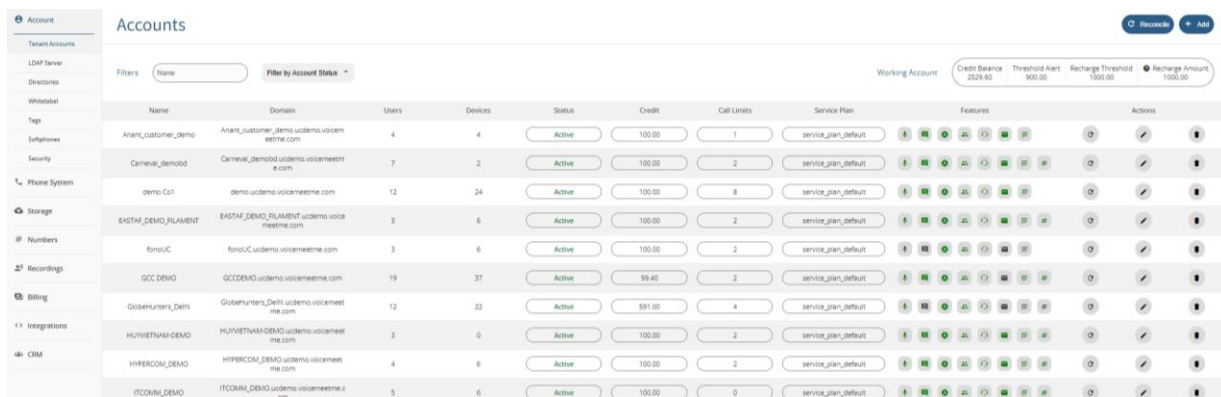
12

Log into the Master Admin Account



The image shows the 'Administration Portal' login interface. It features a white login card on a blue background. The card has a 'Welcome Back!' message with a hand icon, followed by 'Login with your credentials'. There are three input fields: 'Username', 'Password', and 'Account', each with a placeholder text 'Enter your [field name]'. Below these is a 'Sign In' button with a right-pointing arrow. An 'OR' separator is followed by two social login buttons: 'Continue with Google' and 'Continue with Azure'. The version number 'V1.95.12' is visible in the bottom left corner.

- URL: <https://ucdemo.voicemeetme.com/portal/login>
- Credentials
- The Master Account name is “admin”, alternatively use your Reseller account credentials.
- Navigate to Config -> Accounts



The image shows the 'Accounts' management interface. It includes a sidebar with navigation options like 'Tenant Accounts', 'LDAP Server', 'Diagnostics', 'Whitelabel', 'Tags', 'Softphones', 'Security', 'Phone Systems', 'Storage', 'Numbers', 'Recordings', 'Billing', 'Integrations', and 'CRM'. The main area displays a table of accounts with columns for Name, Domain, Users, Devices, Status, Credit, Call Limits, Service Plan, Features, and Actions. A 'Filters' section at the top allows filtering by Name and Account Status. A 'Working Account' summary at the top right shows Credit Balance (2529.60), Threshold Alert (900.00), Recharge Threshold (1000.00), and Recharge Amount (1000.00).

Name	Domain	Users	Devices	Status	Credit	Call Limits	Service Plan	Features	Actions
ANANT_customer_demo	ANANT_customer_demo.ucdemo.voicemeetme.com	4	4	Active	100.00	1	service_plan_default	[Icons]	[Edit] [Delete]
Carneval_demo	Carneval_demo.ucdemo.voicemeetme.com	7	2	Active	100.00	2	service_plan_default	[Icons]	[Edit] [Delete]
demo C01	demo.ucdemo.voicemeetme.com	12	24	Active	100.00	8	service_plan_default	[Icons]	[Edit] [Delete]
EASTAF_DEMO_FLAMENT	EASTAF_DEMO_FLAMENT.ucdemo.voicemeetme.com	3	6	Active	100.00	2	service_plan_default	[Icons]	[Edit] [Delete]
fonuJC	fonuJC.ucdemo.voicemeetme.com	3	6	Active	100.00	2	service_plan_default	[Icons]	[Edit] [Delete]
GCC DEMO	GCCDEMO.ucdemo.voicemeetme.com	19	37	Active	99.40	2	service_plan_default	[Icons]	[Edit] [Delete]
GlobeHunters_DEM	GlobeHunters_DEM.ucdemo.voicemeetme.com	12	22	Active	591.00	4	service_plan_default	[Icons]	[Edit] [Delete]
HUYVIETNAM_DEMO	HUYVIETNAM_DEMO.ucdemo.voicemeetme.com	3	0	Active	100.00	2	service_plan_default	[Icons]	[Edit] [Delete]
HYPERCOM_DEMO	HYPERCOM_DEMO.ucdemo.voicemeetme.com	4	6	Active	100.00	2	service_plan_default	[Icons]	[Edit] [Delete]
ITCOMM_DEMO	ITCOMM_DEMO.ucdemo.voicemeetme.com	5	6	Active	100.00	0	service_plan_default	[Icons]	[Edit] [Delete]

Select  on Add New Account

Click on Add New Account to create a new account.

Specify account Name, unique Realm and Administrator credentials

Add Account

Set Up

General Account

Account Name Realm Is Reseller ☐ Timezone

Contact Information

Technical

Email Name Number

Company Address

Country Region Locality Street Address ZIP Code

Service Plan

Service Plan

(+) Override Service Plan

Collaboration

State

Enabled ☐

Auto Provisioning

UCP

Enabled ☒

Select Audio Codes

Call Recording

State

Enabled ☐

Call Restrictions

Limits

Twonky trunks

Inbound trunks

Outbound trunks

Features

Teams Calling ☐

UCP Mobile

Enabled ☒

Select Audio Codes

Select Video Codes

Config

Retention Duration

Credit

Add Recharge Threshold Max Daily Fraud Risk

Threshold Alert Recharge Amount

Account Name	Descriptive account name.
Account Realm	Used as part of SIP Registration credentials and provides isolation between different accounts, allowing one to have the same extension number (eg: 1000) on two different accounts. The realm should be defined as: {account name}.domain.com eg: bestbuy.cloudbpx.com

Timezone	Set the timezone of the customer. Each account can have a different time zone.
Administrator	For security reasons let the system define the password which will be emailed to the administrator's email.

Specify the Service Plan

Service Plan is used to define how and what the customer should be billed for. Almost every feature of the platform can be “charged for”: the end customer will be prompted to accept new charges when adding new features or creating resources such as users and devices.

Service plan can be overridden via the + sign. Admin can choose to provide special per user, DID or porting pricing or offer a discount from the MSRP. Furthermore, Admin can setup multiple Service Plans that are tied to custom Rate Decks that are used for per-minute call rating and billing.

Specify Limits

Specify the number of consecutive two way calls this account can make and disable post pay and enable pre-pay for metered calls.

The pre-pay feature is used as a part for fraud prevention. Each account can spend a maximum amount of money per-month on per-minute (long distance) calls, thus in the event of fraud, the customer is protected by a “soft cap”. The account will self top-up once during a month in order to minimize service disruption due to normal calling.

Add Phone Numbers to the New Account

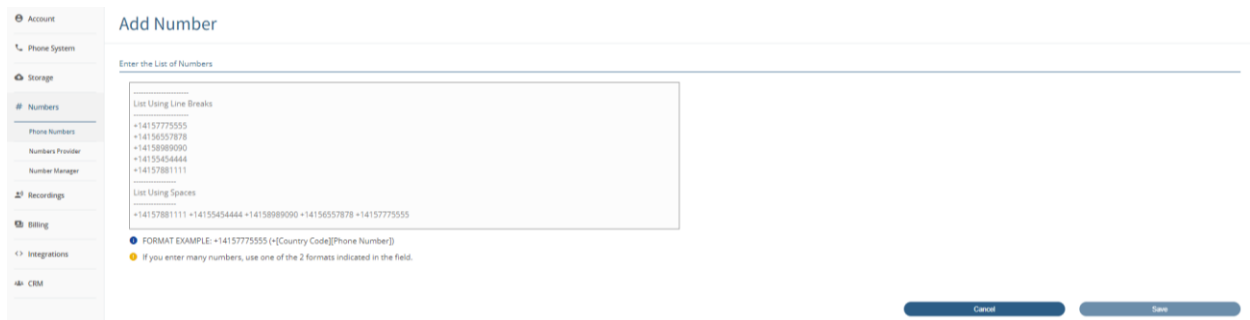
Log into the Portal

- <https://ucdemo.voicemeetme.com/portal/configuration/numbers/general>
- Navigate to Config -> Numbers



Find the an account and click on

Add Numbers



Add new numbers to the account: make sure to use the e.164 notation: for USA and Canada +14164445555, for other countries use the country code +622129303030.

At this point the phone numbers have been assigned to the customer's account. The next step is to log into the customer's account and configure the cloud PBX.

Set Account level Caller ID

- Navigate to SmartPBX from Portal
- Option1
 - Click on CallerID in Dashboard Section of SmartPBX
- Option2
 - Select Callflow Application in SmartPBX
 - Select Account Settings
 - Set callerid for normal calls
 - Set callerid for emergency calls
- Note: one must have at least one phone number bound to the tenant account to set account level callerid

Set Account level Music on Hold

- Navigate to SmartPBX from Portal
- Option1
 - Click on Hold Music in Dashboard Section of SmartPBX
- Option2
 - Select Callflow Application in SmartPBX
 - Select Play Media
 - Upload Media files in MP3 format
 - Specify the name of uploaded MP3 file and click Save

The screenshot displays the 'Play Media' section of the SmartPBX interface. On the left is a sidebar with a search bar and an 'Add' button. The main content area is titled 'Play Media' and includes a 'Return to Callflows Home' link. Below this is a 'Create Media' section with two tabs: 'Basic' (selected) and 'Advanced'. The 'Basic' tab contains a form with three fields: 'Name' (text input), 'Media Type' (dropdown menu set to 'File'), and 'File Input' (a 'Choose File' button with the text 'No file chosen' below it). A green 'Save' button is located at the bottom right of the form.

Portal Account Administration

Admin portal allows further account configuration and feature selection. Navigate to <https://<domain>/portal> and login with admin credentials.

Account	Name	Domain	Users	Devices	Status	Credit	Call Limits	Service Plan	Features	Actions
Tenant Accounts	Anant_customer_demo	Anant_customer_demo.voicemeet.com	4	4	Active	100.00	1	service_plan_default	[Icons]	[Edit, Delete, Add]
LDAP Server	Carneval_demo	Carneval_demo.voicemeet.com	7	2	Active	100.00	2	service_plan_default	[Icons]	[Edit, Delete, Add]
Directories	demo-Cst	demo.voicemeet.com	12	24	Active	100.00	8	service_plan_default	[Icons]	[Edit, Delete, Add]
Whiteboard	EASTAF_DEMO_PLAMENT	EASTAF_DEMO_PLAMENT.voicemeet.com	3	6	Active	100.00	2	service_plan_default	[Icons]	[Edit, Delete, Add]
Tags	fonuUC	fonuUC.voicemeet.com	3	6	Active	100.00	2	service_plan_default	[Icons]	[Edit, Delete, Add]
SIPphones	GCC DEMO	GCC DEMO.voicemeet.com	19	37	Active	99.40	2	service_plan_default	[Icons]	[Edit, Delete, Add]
Security	GlobeHunters_Demo	GlobeHunters_Demo.voicemeet.com	12	22	Active	99.00	4	service_plan_default	[Icons]	[Edit, Delete, Add]
Phone System	HUVIETNAM DEMO	HUVIETNAM DEMO.voicemeet.com	3	0	Active	100.00	2	service_plan_default	[Icons]	[Edit, Delete, Add]
Storage	HYPERCOM DEMO	HYPERCOM DEMO.voicemeet.com	4	6	Active	100.00	2	service_plan_default	[Icons]	[Edit, Delete, Add]
# Numbers	ITCOMM DEMO	ITCOMM DEMO.voicemeet.com	5	6	Active	100.00	0	service_plan_default	[Icons]	[Edit, Delete, Add]
Recordings										
Billing										
Integrations										
CRM										

Administrators can view and/or edit accounts on the platform including number of users, devices, credit, limits and features.

Feature	Description
Credit	Ability to set daily Fraud credit for rated calls
Call Limits	Set the concurrent call limits per account
Call Recording	Set call recording limits
Collaboration	Enable collaboration for the account
Auto provisioning	Enable Desktop and or Mobile Apps per account
Meetings	Enable Video Conferencing features per account

Add Users to the New Account

Admin can add a single or bulk upload users to an account using a CSV document. Navigate to Portal -> Users and then click Add.

Users

Reports

7 Users

Filter by Name

Filter by Email

Filter by Extension

Filter by Number

Status

Invitation

Role

Filter by Tags

	Enabled	Name	Email	Extension	Role	Tags	Numbers	Cellphones	Get Started Link	Invitation	Actions
☐	☒	Noor Sakib	noor.sakib@ssd-tech.io	1005	User				GO	Pending	Edit Delete
☐	☒	Rizwan ssd	rizwan@ssd-tech.io	1002	User				GO	Pending	Edit Delete
☐	☒	Readur Rahman	readur.rahman@ssd-tech.io	1004	User				GO	Pending	Edit Delete
☐	☒	Matin Carneval	matin@dotimes.com.sg	1001	Admin Total Access				GO	Accepted	Edit Delete
☐	☒	Mahbub ssd	mahbub@ssd-tech.io	1003	Admin Total Access				GO	Pending	Edit Delete
☐	☒	Test Admin	mohit@multicommm.com	2001	User				GO	Not Invited	Edit Delete
☐	☒	Test Admin2	aveek@multicommm.com	2002	Admin Total Access				GO	Pending	Edit Delete

Users

Reports

Add User

Personal

First Name

Last Name

Email

Extension

Password

Confirm Password

Invite to Platform

Access Level

Level

Restricted by tags

Operator

Tags

Callcenter

Supervisor

Conferencing

Enabled

Company Directory

Include in Directory

Cancel

Save User

- Specify First or Last Name
- Email
- Role: User or Admin
 - User cannot log into the portal
 - Admin can log into the portal and will have certain permissions based on Admin Type
- Admin Type
 - Total access - full admin
 - Read only allows user to read only, not able to configure or change anything
 - User activation feature allows billing department to enable or disable users only
 - Call Center supervisor is allowed to only manipulates the call center queues
- Extension
- Invite to platform allows one to auto invite user to platform via email on creation
- Conferencing allows admin to setup a audio conference for the user
- If user is included in directory, user will show up in Desktop, Mobile and IP phones with presence

Invite Users to the platform

Admin can invite a single user or all users to the platform via Email or SMS. Users will receive an onboarding link that can be used to auto login to Web, Desktop or Mobile phone. There is no need to share passwords or create complicated onboarding procedures.

Users

Reports

7 Users

Filter by Name

Filter by Email

Filter by Extension

Filter by Number

Status

Invitation

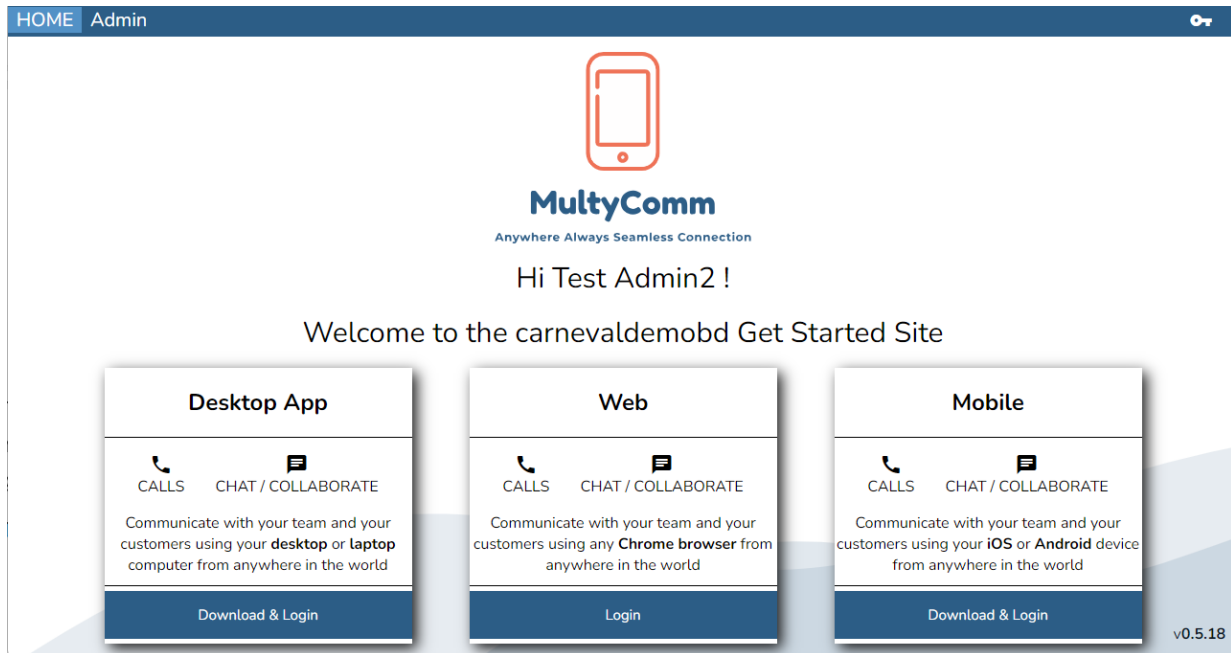
Role

Filter by Tags

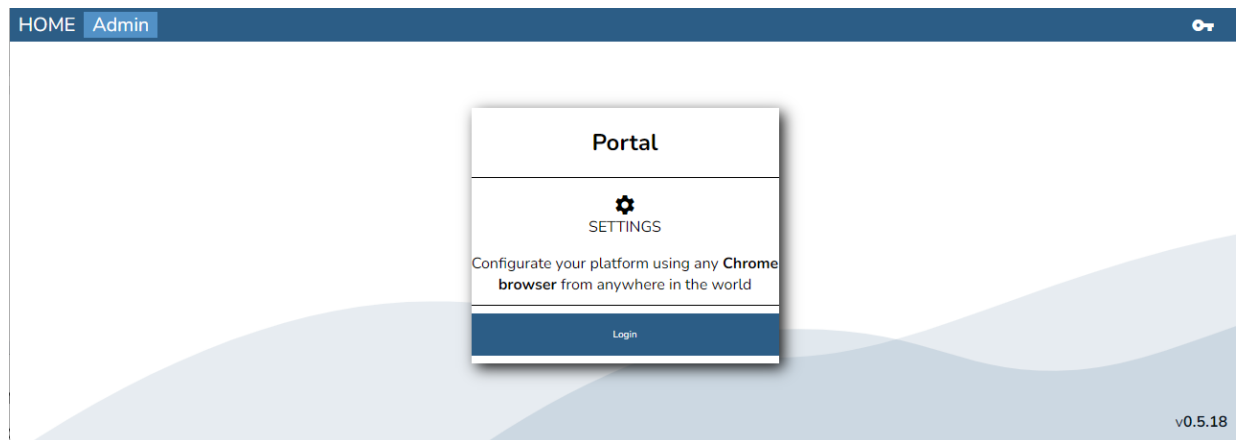
	Enabled	Name	Email	Extension	Role	Tags	Numbers	Cellphones	Get Started Link	Invitation	Actions
☐		Noor Sakib	noor.sakib@ssd-tech.io	1005	User					Pending	
☐		Rizwan ssd	rizwan@ssd-tech.io	1002	User					Pending	
☐		Readur Rahman	readur.rahman@ssd-tech.io	1004	User					Pending	
☐		Matin Carneval	matin@dotlines.com.sg	1001	Admin Total Access					Accepted	
☐		Mahbub ssd	mahbub@ssd-tech.io	1003	Admin Total Access					Pending	
☐		Test Admin	mohit@multicommc.om	2001	User					Not Invited	
☐		Test Admin2	aveek@multicommc.om	2002	Admin Total Access					Pending	

Welcome Invitation

When a user receives the email and clicks the onboarding link, they will be greeted with a personal message and options to auto login to a Desktop app, Web app or Mobile app. Screen will change if the link is opened on Desktop or Mobile.



If the invited user is admin, the Admin section in the top bar, will also lead the Admin to the Portal magic link.



Users can also change their password in the top right hand corner section of the onboarding link. Via the  icon.

Configure a new Cloud PBX

In this section we will outline all the steps necessary to configure a new customer cloud PBX.²

Summary

1. Log into the Master Admin Account
2. Select Account Management
3. Select the Account from the account list.
4. Log in to the account using the “Use Account” button.
5. Select Smart PBX application
6. Configure Main Number
 - a. Bind “spare number” to the Main Number
 - b. Configure working hours
 - c. Configure holiday dates
 - d. Configure Main Number Route / IVR
 - i. Upload IVR greeting
 - e. Bind “spare number” to the Conference
 - i. Upload IVR greeting
 - f. Bind “spare number” and an email to the FaxBox
7. Create Users (if not already created in portal above)
 - a. Bind extension to user
 - b. Bind DID to user
 - c. Bind a Device to user
 - i. SIP Phone
 - ii. Cell Phone
 - d. Configure extension features
 - i. Find me follow me
 - ii. Voicemail
 - iii. Faxbox
8. Create Groups
 - a. Eg: Sales, Tech Support

² This section assumes that the Multycomm team has already installed and preconfigured the Multycomm cluster with Service Plans, Rate Decks, Carrier Gateways and confirmed that all sections of the Multycomm system are operational and functioning correctly.

Log into the customer's account

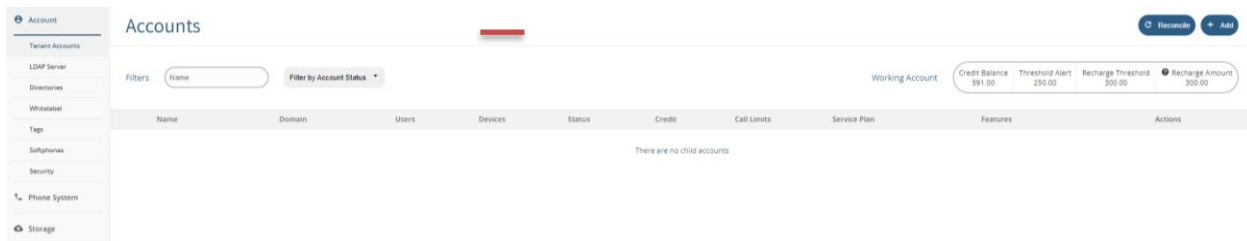
Log into the newly created account by selecting the “Use Account” button.



Master Admin can “masquerade” into any sub-account.

TIP: Current logged in account name is displayed in the top right corner.

Once the admin is logged into the customer's account, click on the SmartPBX Icon to login into the SmartPBX application.

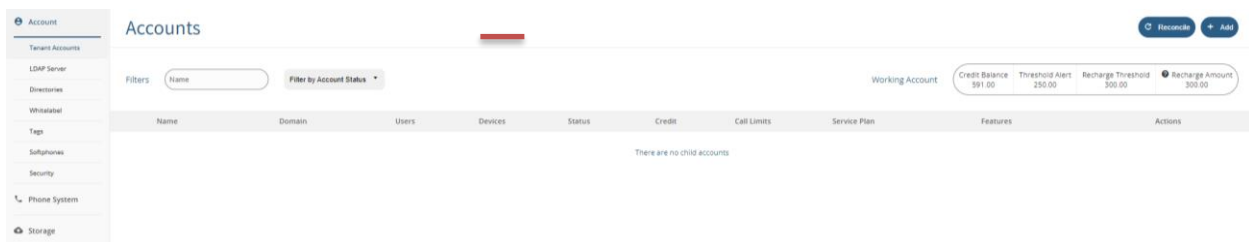


Smart PBX Configuration

The Smart PBX app combines the most common PBX functionality with a beautiful user interface and an intuitive experience. An advanced set of features such as individual fax-to-email boxes, call forwarding, and hot desking are easy to understand and utilize. A simple dashboard provides a quick but comprehensive overview of system configuration and status. Within this dashboard, review total users and devices, assign conference bridges and phone numbers. Create customized greetings and call routes to give your business that professional touch. Set functionality with a beautiful user interface and an intuitive experience. Set up a main business phone number and utilize the pre-built Virtual Receptionist to handle inbound calls.

To go into Select the Smart PBX application from the application exchange page.

Select the Smart PBX application from the portal top bar.

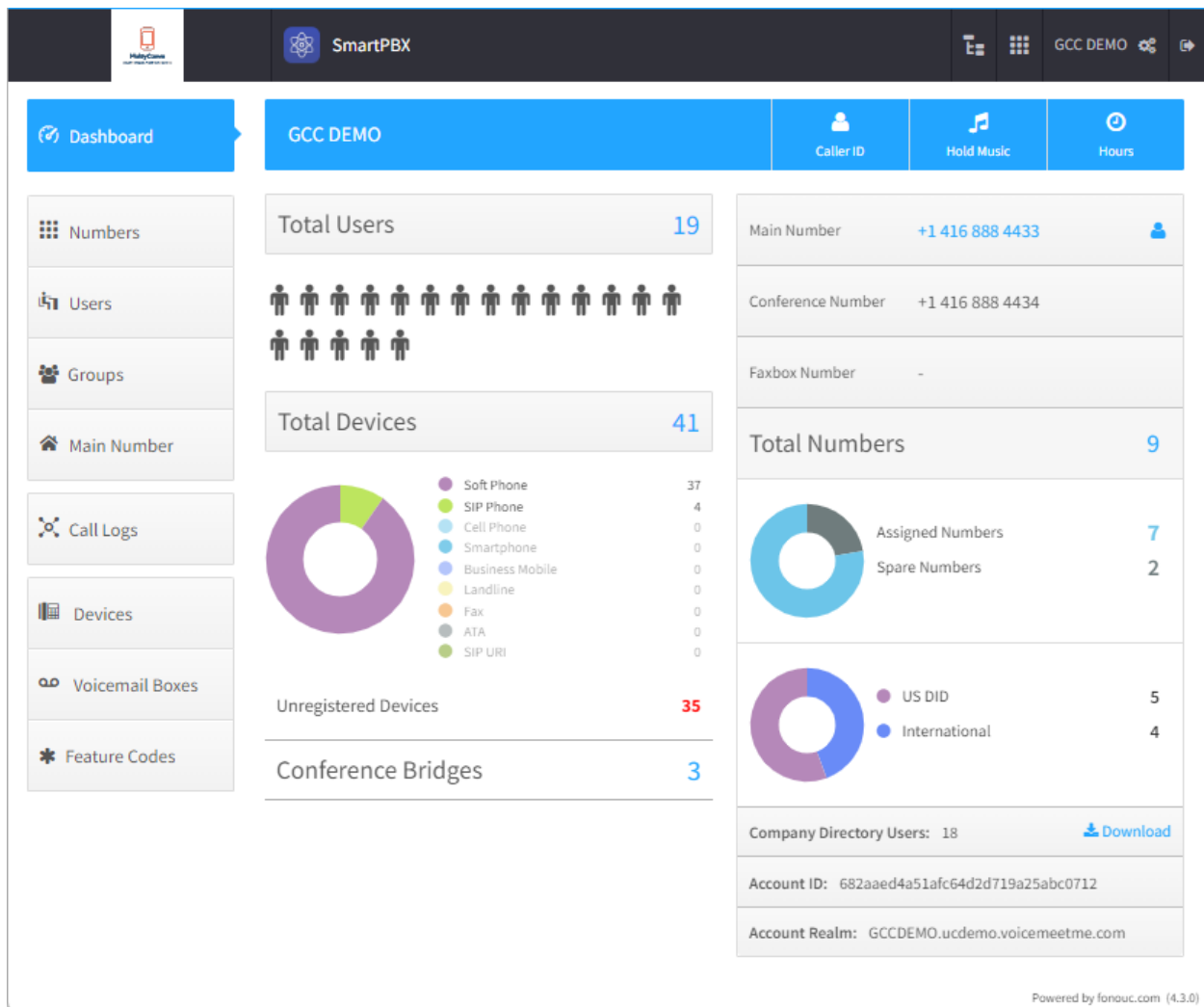


The screenshot shows the 'Accounts' management page. On the left is a sidebar with navigation links: Account, Tenant Accounts, LDAP Server, Directories, Whitelist, Tags, Softphones, Security, Phone System, and Storage. The main header area includes the title 'Accounts', a search filter 'Name', a dropdown 'Filter by Account Status', and a 'Records' button. Below this is a summary row for the 'Working Account' showing 'Credit Balance: 991.00', 'Threshold Alert: 250.00', 'Recharge Threshold: 300.00', and 'Recharge Amount: 300.00'. The main content area is a table with columns: Name, Domain, Users, Devices, Status, Credit, Call Limits, Service Plan, Features, and Actions. The table is currently empty, displaying the message 'There are no child accounts'.

Name	Domain	Users	Devices	Status	Credit	Call Limits	Service Plan	Features	Actions
There are no child accounts									

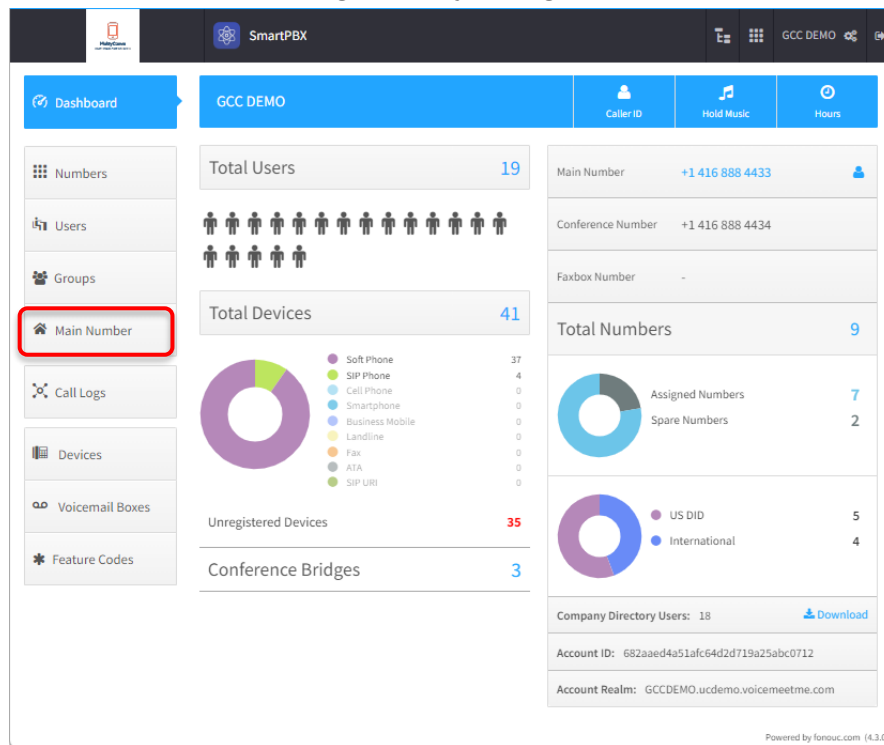
SmartPBX Dashboard

When you click on the Smart PBX application it automatically defaults to the dashboard. The Dashboard consists of the whole overview of your organizational structure on the Multycomm PBX system. This app is used most for your configuration and setup for your Numbers, Users, Groups, Main Numbers, Devices, Voicemail Boxes, Feature Codes, Caller ID, Hold Music and Hours of operation. You are able to even view Call Log files for all users within your organization. You can click on each main heading within the dashboard or on the side panel to access each section of the Smart PBX application. The most frequently used features are prioritized, creating a more efficient work environment.

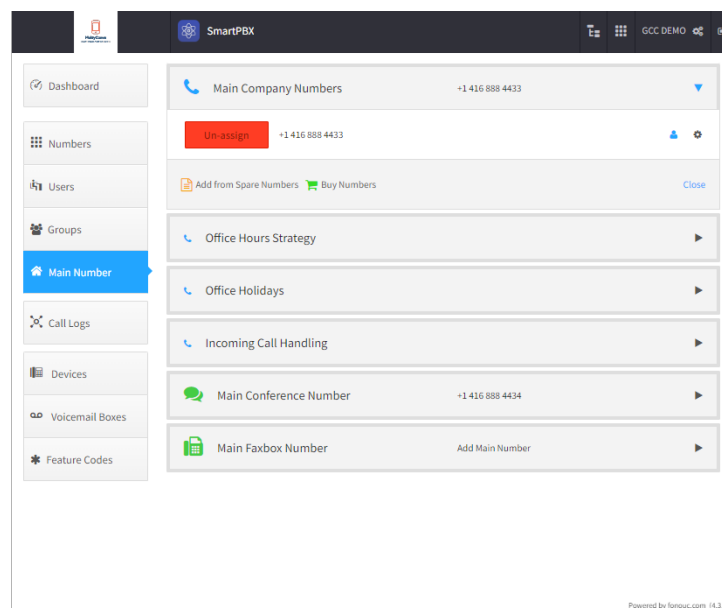


Main PBX Number

Start customer cloud PBX configuration by adding a Main Number to the account.³



Select “Add from Spare Numbers” and bind the Main number to the PBX. An incoming call to that number will be routed to the PBX IVR.



³ The Master Admin should have already added numbers/DIDs to the account via the Number Manager.

Specify the hours of operation. This option will allow one to configure business hours and closed hours IVR.

SmartPBX

Dashboard

Numbers

Users

Groups

Main Number

Call Logs

Devices

Voicemail Boxes

Feature Codes

Main Company Numbers +1 416 888 4433

Office Hours Strategy

☐ 24 hours Open Office
Choose this option if you want incoming calls to be handled the same way all the time.

☒ Custom Office Hours
Choose this option if you want incoming calls to be handled differently when your office is closed.

Custom Hours Company Timezone: Muscat (Asia)

☒ Monday	9:00	to	17:00	Open
☒ Tuesday	9:00	to	17:00	Open
☒ Wednesday	9:00	to	17:00	Open
☒ Thursday	9:00	to	17:00	Open
☒ Friday	9:00	to	17:00	Open
☐ Saturday				Closed
☐ Sunday				Closed

☐ Are you closed for lunch hours?

PBX Inbound Routing

Specify the PBX incoming call routes: The SmartPBX attempts to provide a simplified IVR configuration. If a more complicated IVR is required, then the admin must use the Callflow application.

Select “Virtual Receptionist” to bind a Media file to the IVR.

SmartPBX

Dashboard

Numbers

Users

Groups

Main Number

Call Logs

Devices

Voicemail Boxes

Feature Codes

Main Company Numbers +1 416 888 4433

Office Hours Strategy

Office Holidays

Incoming Call Handling

Open Hours Holidays

Select the way you want your callers to reach you when your office is Open.

Incoming Call

↓

Virtual Receptionist

Incoming Call

↓

Users

abhishek+ Dixit

↓

Virtual Receptionist

Incoming Call

↓

Users

abhishek+ Dixit

↓

Voicemail

abhishek Dixit's VM...

Cancel Save Changes

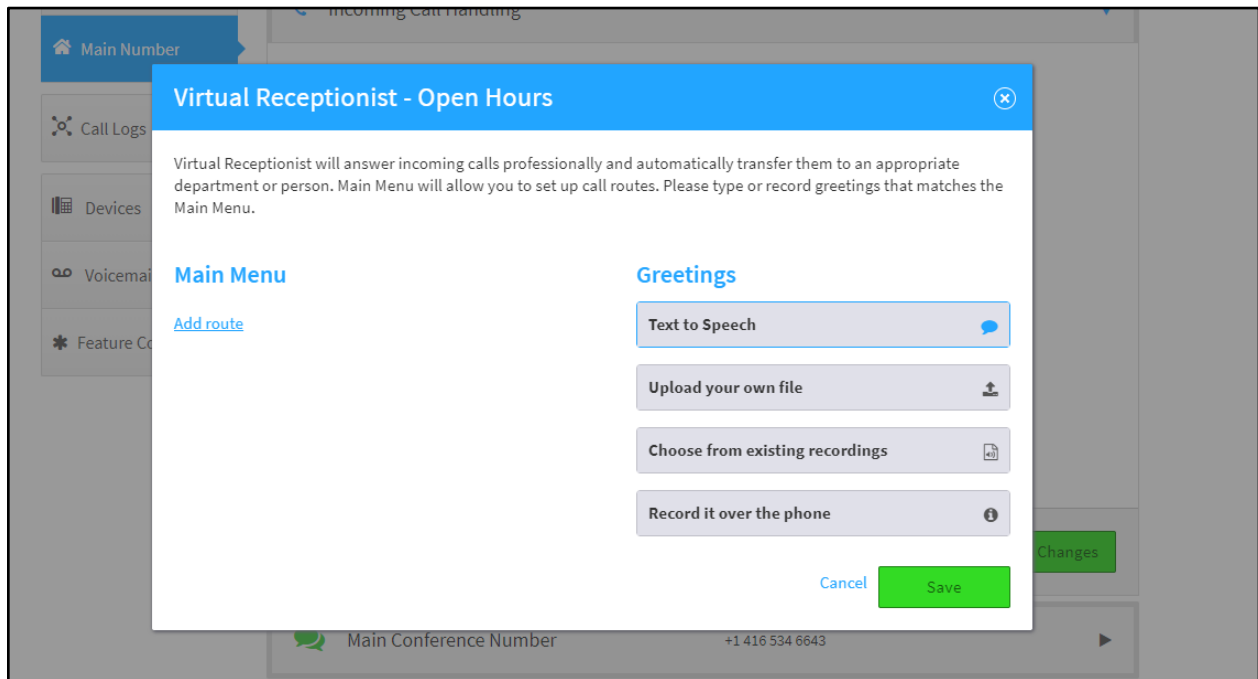
Main Conference Number +1 416 888 4434

Main Faxbox Number Add Main Number

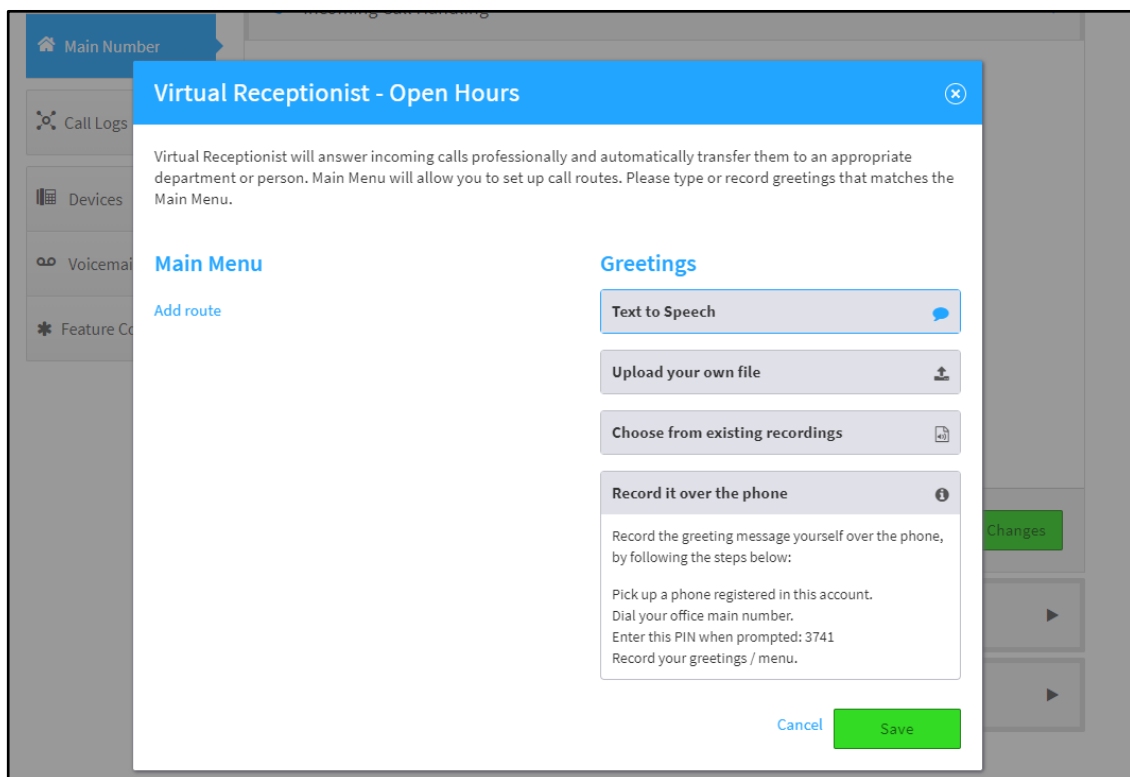
Powered by fonou.com (4.3.0)

The Virtual Receptionist plays a greeting and requests the incoming user to select an action from a menu. Eg: Press 1 for Sales, 2 for Technical support or if you know the extension dial it.

Click on Upload to upload an MP3 or WAV greeting.



Alternatively admin can dial in and record the Main Greeting.



Conference

Main conference requires a phone number/DID. Click on “Add from spare numbers” and bind a phone number to the main phone conference. When customers dial this number, they will be presented with a greeting and will be asked to enter a conference room and a pin code.

The screenshot displays the SmartPBX web interface. The top navigation bar includes the SmartPBX logo, a user profile icon, and a 'GCC DEMO' label. The left sidebar contains a menu with options: Dashboard, Numbers, Users, Groups, Main Number (highlighted in blue), Call Logs, Devices, Voicemail Boxes, and Feature Codes. The main content area shows the configuration for the 'Main Number'. It includes a list of numbers with their corresponding phone icons and a right arrow. The 'Main Conference Number' is highlighted with a green speech bubble icon and a blue downward arrow. Below this, there is a red 'Un-assign' button and the number '+1 416 888 4434'. At the bottom, there is a section for 'Main Faxbox Number' with an 'Add Main Number' button. The interface also features a 'Custom Greeting' section with a speaker icon, and buttons for 'Add from Spare Numbers', 'Buy Numbers', and 'Close'.

Note: Individual conferences are configured under User configuration.

Faxbox

Main Faxbox requires a phone number/DID. Click on “Add from spare numbers” and bind a phone number to the main faxbox. When customers send faxes to the faxbox phone number/DID, the fax will be converted to a Tiff/PDF and sent via email to the PBX admin or individual user. Furthermore, PBX users can email a PDF to a special Multycomm faxbox email account that will transmit the fax via PSTN to the remote fax machine and email back the result.

The screenshot displays the SmartPBX web interface. The top navigation bar includes the MultyComm logo, the SmartPBX title, and a GCC DEMO button. The left sidebar contains a menu with options: Dashboard, Numbers, Users, Groups, Main Number (highlighted in blue), Call Logs, Devices, Voicemail Boxes, and Feature Codes. The main content area shows the configuration for the Main Number. It includes a list of settings: Main Company Numbers (+1 416 888 4433), Office Hours Strategy, Office Holidays, Incoming Call Handling, Main Conference Number (+1 416 888 4434), and Main Faxbox Number (Add Main Number). A red message states "Please add a main faxbox number." At the bottom, there are buttons for "Add from Spare Numbers" and "Buy Numbers", and a "Close" button. The footer indicates the system is powered by fonouc.com (4.3.0).

Individual faxbox configuration is performed under per User configuration.

User Creation

Select “Users” from the SmartPBX dashboard and click on “Add User”. Once the user is created, the credentials will automatically be emailed to the user and the system administrator.

The screenshot shows the SmartPBX dashboard interface. On the left is a sidebar with navigation links: Dashboard, Numbers, Users (highlighted), Groups, Main Number, Call Logs, Devices, Voicemail Boxes, and Feature Codes. The top of the dashboard shows 'Total Users 25' and an 'Add User' button. A search bar is located in the top right corner. The main content area displays a table of users with the following columns: Users Settings, Extensions, Phone Numbers, Devices, and User Features. The table lists ten users with their respective details.

Users Settings	Extensions	Phone Numbers	Devices	User Features
Anthony Powell	1007	None	 	 
Billy Bryant	1023	None	 	
Carol Patel	1018	None	 	
Donna Robertson	1006	None	 	
Harold Watson	1017	None	 	
Harry Knight	1015	None	 	
Henry Little	1010	None	 	
Jack Scott	1009	None	 	
Jane Hanson	1024	None	 	

Security: Always let the system define and email a random password to the end user.

Please note: A user can be an individual or a device. Example a conference room with a phone will need a user account. All extensions are assigned to a user or group.

User PBX Features

User PBX feature section allows the administrator to configure per user features such as: find me follow me, faxbox, voicemail, conferences etc.

The screenshot displays the 'User PBX Features' configuration page for a user named Anthony Powell. The interface includes a sidebar with navigation options: Dashboard, Numbers, Users (selected), Groups, Main Number, Call Logs, Devices, Voicemail Boxes, and Feature Codes. The main content area shows a summary of the user's settings across five tabs: Users Settings, Extensions, Phone Numbers, Devices, and User Features. The 'Users Settings' tab is active, showing a grid of feature toggles for Anthony Powell (extension 1007). The features and their status are as follows:

Feature	Status
Caller-ID Number	Off
Call Forwarding	Off
Hot-Desking	Off
Voicemails	On
Faxbox	Off
Conference Bridge	On
Find me, Follow me	Off
Music-On-Hold	Off
Customized Call Recording	Off
Do Not Disturb	Off

Caller ID

If enabled in numbers this feature can be turned on or off here. If you don't enable this feature, the Caller-ID displayed by default will be the one configured on the Main Number of this account.

Caller-ID Number

 User Caller-ID Number Settings ☒

Show my Caller-Id Number as

 If you don't enable this feature, the Caller-ID displayed by default will be the one configured on the Main Number of this account.

[Cancel](#) [Save Changes](#)

Call Forwarding

To activate this feature click on On. You can specify what device or number you would like the calls forwarded to. You can choose to leave voicemails on the forwarded number, Forward direct calls only and Keep the original Caller-ID. Please remember to save changes.

Forwarding can be enabled from SmartPBX or from the Web/Desktop Softphone.

The screenshot shows a web interface for user settings. At the top, there are tabs: "Users Settings", "Extensions", "Phone Numbers", "Devices", and "User Features". Under "Users Settings", the user "Anthony Powell" is selected, with extension "1007" and "None" phone numbers. Below this, there are three main settings: "Caller-ID Number", "Call Forwarding", and "Hot-Desking". The "Call Forwarding" option is highlighted with a red underline. A modal dialog titled "Call Forwarding" is open in the foreground. It has a blue header with a close button. Inside the dialog, there is a "User Call Forwarding Settings" section with a circular icon and three buttons: "Off", "Failover Mode", and "On". The "On" button is highlighted with a red underline. Below this, there are two input fields: "Forward all calls to" with a dropdown menu showing "A Desk Phone", and "This number" with a text input showing "+14157283992". At the bottom of the dialog, there are three checkboxes: "Leave voicemails on forwarded numbers" (unchecked), "Forward direct calls only" (unchecked), and "Keep Original Caller-ID" (checked). At the very bottom of the dialog are two buttons: "Cancel" and "Save Changes".

Users Settings Extensions Phone Numbers Devices User Features

Anthony Powell 1007 None

Caller-ID Number Call Forwarding Hot-Desking

Call Forwarding

User Call Forwarding Settings Off Failover Mode On

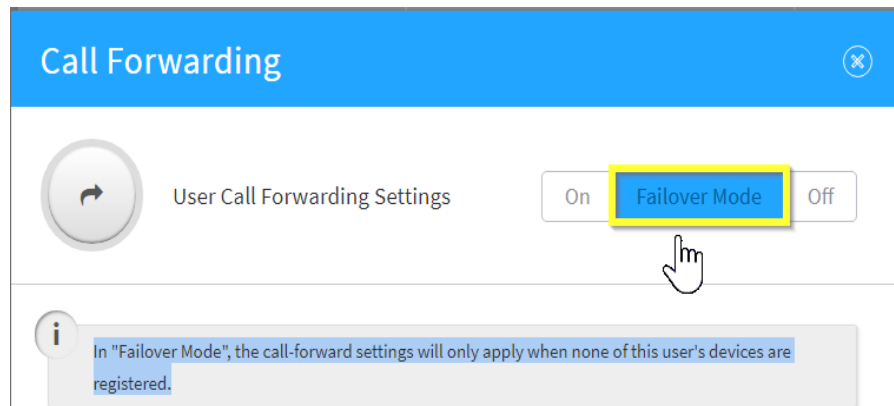
Forward all calls to A Desk Phone

This number +14157283992

- ☐ Leave voicemails on forwarded numbers
- ☐ Forward direct calls only
- ☒ Keep Original Caller-ID

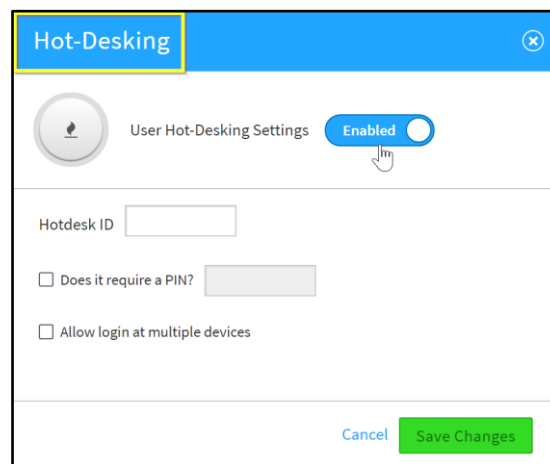
Cancel Save Changes

You can set Call Forwarding as part "**Failover Mode**", the call-forward settings will only apply when none of this user's devices are registered.

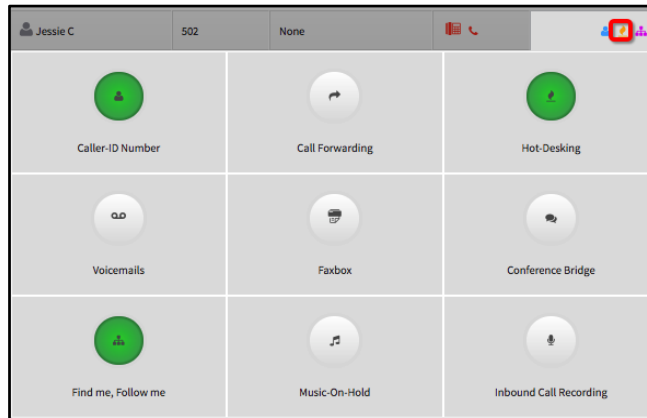


Hot-Desking

This feature enables multiple users to log in and make calls on a single PBX enabled handset using their four-digit operator extension number and PIN. Example uses for Hot-Desking include: Multiple employees using 1 phone at different times during the day, different employees assuming reception responsibilities, unassigned desks where employees play musical chairs, or running multiple entities from a single device.



1. Enable hot-desking for a specific User A
2. Enter a Hotdesk ID (Must Be Unique)
3. For extra security you can assign the Hot-Desk a pin number but this is not required
4. Checking the Allow login at multiple devices allows multiple logins for employees
5. Press Save Changes and **Hot-Desking** will be enabled
6. **Important:** Find-me-Follow me must be **disabled** for User A.
 - a. If enabled, incoming Hotdesk calls will not work and all incoming calls to User A will continue to be sent to User A and not the Hotdesk user device.



Hotdesk Login

- From User B phone, simply dial the feature code *11
- Enter in the Hot-Desk ID (and PIN) of User A (the user we want to take over calls from).
 - Now all outgoing calls from User B will use User A caller id
 - All inbound calls to User A will be routed to both User A devices and Hotdesk User B devices
 - Hotdesk User B is able to answer all User A calls
- Note: the above process can be repeated for multiple Users thus, User B can take over phone control for multiple Users (e: User A and User C) via their unique Hot-Desk ID.

Warning: However, if User B takes over multiple hotdesk phones, outbound caller id will be broken/non-deterministic when User B makes outbound calls as there is no way to define which hotdesk callerid User B should use.

Hotdesk Logout

- Hotdesk User B can logout from all hotdesk devices by dialing *12
- Alternatively User B can also dial *13 and choose which hotdesk device to logout from.

Voicemails

If you have set up voicemails for each user you will be able to enable or disable them voicemail under each user or check to have them delete messages from the database after sending notifications. All other Voicemail setup will be explained in section Voicemail Box

Voicemails



Voicemail-to-Email Settings

Enabled

An e-mail will be sent to `edgard.cuadra@startelecom.ca` for every new voicemail

☐ Delete messages from database after sending the notification.





If you want to change where email notifications go to, please edit the user's email address in the User Administration section

Cancel

Save Changes

38

Faxbox

To enable user to receive an email when a fax comes in can be done in the Faxbox feature

Faxbox

 Faxbox ☒ Enabled

Fax to Email

Faxes sent to will be converted to emails and sent to edgard.cuadra@startelecom.ca.

Email to Fax

Emails sent to **{ destination number }@startel.starpbx.ca** with a PDF (or TIFF) attachment will fax the attachment to **{ destination number }**. The email must come from edgard.cuadra@startelecom.ca.

[Show help](#)

[Cancel](#) [Save Changes](#)

Conference Bridge

You can create a conference call by providing a conferencing phone number and your personal room number to the participants this is enabled under the user.

Conference Bridge



Personal Conference Bridge

Enabled



Conferencing Phone Number(s)

+1 877 398 3016

+1 647 729 8386

Personal Conference Room number



You can create a conference call by providing a conferencing phone number and your personal room number to the participants

Cancel

Save Changes

Find Me, Follow Me

Always be available with this Multycomm PBX feature. Find Me, Follow Me enables incoming calls to “find me” and deliver those calls to any location, while the “follow me” provides the ability to receive calls at any number of your designated devices. Your “follow me” phone number is routed through a list of numbers you choose (cell, home, office, etc.), in your preferred order, until the call connects to you. If incoming calls are not answered within a specific number of rings, they are instantly transferred to the voicemail system. Callers are prompted to leave voice messages, which can be retrieved later, at your convenience. You also have the option for Do Not Ring on any particular device.



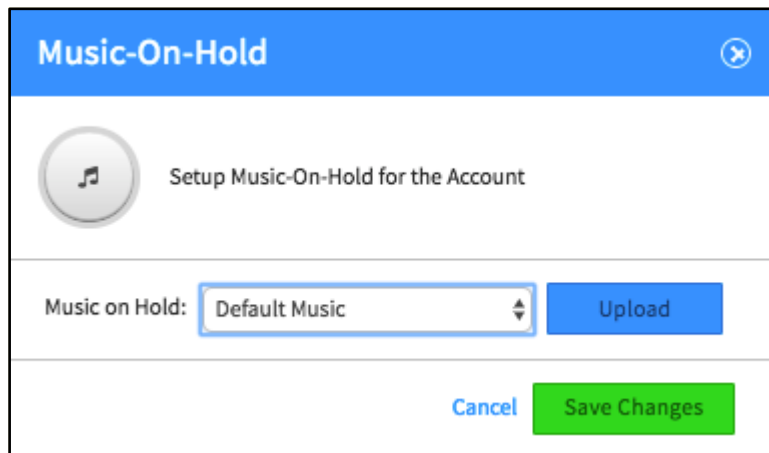
The interface for the 'Find me, Follow me' settings is shown. It features a blue header with the title 'Find me, Follow me' and a close button. Below the header, there is a section for 'Find me, Follow me Settings' with an 'Enabled' toggle switch. A 'Distribute' button is located to the left of a timeline. The timeline shows a sequence of devices to be called: 'Michael's SIP ...' and 'Michael's Cell ...'. The timeline is marked with time intervals: 0 Sec, 20 Sec, 40 Sec, 60 Sec, 80 Sec, 100 Sec, and 120 Sec. The 'Do not ring' option is also present. The 'Save Changes' button is highlighted in green.

Device	Start Time (Sec)	End Time (Sec)	Do not ring
Michael's SIP ...	0	20	☐
Michael's Cell ...	14	40	☐

You are able to set the distribution of how long to wait before you try the next device

Music on Hold

Here is where you can set Music on Hold for individual users , If non is set then it will use the default



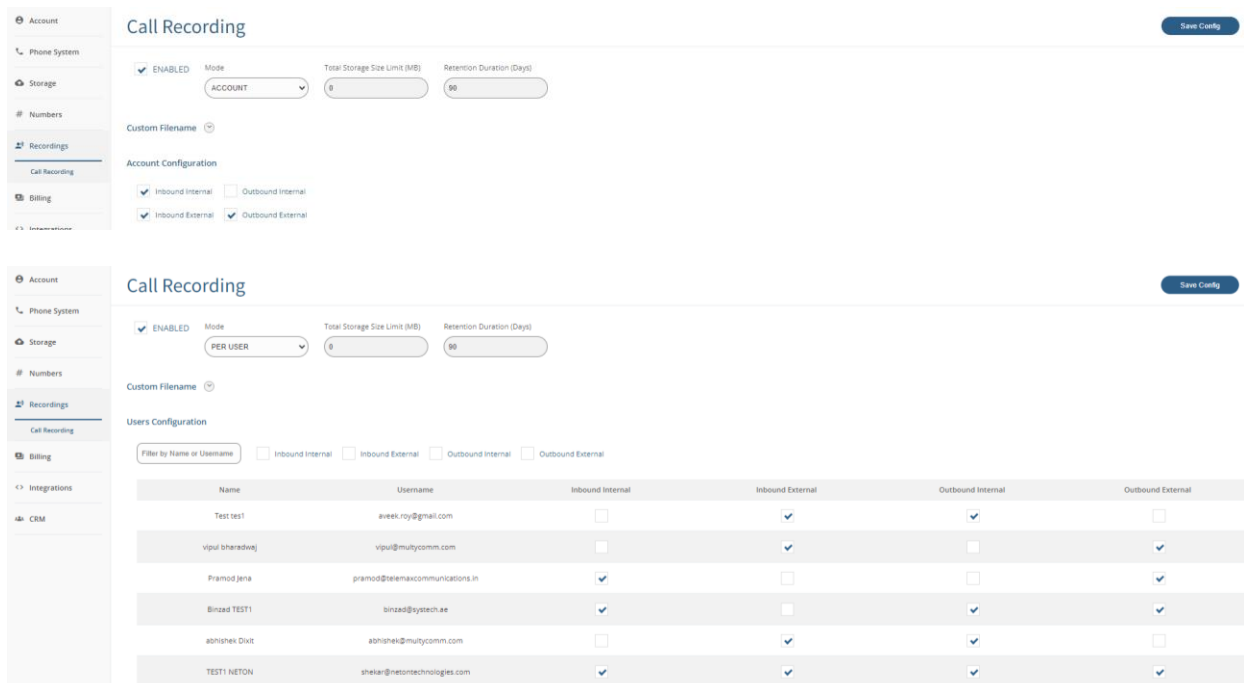
Music on Hold for Main Number

From this window you have the choice of our Default Music, or you can upload your own file and use it. Compatible formats are .WAV, .MP3, and .WMA

Customized Call Recording

Call recording can be enabled per Account or per User in Portal.

Navigate to Portal -> Config -> Call Recording



Call Recording Configuration - Account Mode

ENABLED Mode: ACCOUNT Total Storage Size Limit (MB): 0 Retention Duration (Days): 90

Custom Filename: [Dropdown]

Account Configuration

- ☒ Inbound Internal ☐ Outbound Internal
- ☒ Inbound External ☒ Outbound External

Call Recording Configuration - User Mode

ENABLED Mode: PER USER Total Storage Size Limit (MB): 0 Retention Duration (Days): 90

Custom Filename: [Dropdown]

Users Configuration

Filter by Name or Username ☐ Inbound Internal ☐ Inbound External ☐ Outbound Internal ☐ Outbound External

Name	Username	Inbound Internal	Inbound External	Outbound Internal	Outbound External
Test test	aveek.roy@gmail.com	☐	☒	☒	☐
vipul bharadwaj	vipul@multicom.com	☐	☒	☐	☒
Pramod jena	pramod@telemaxcommunications.in	☒	☐	☐	☒
Binzad TEST1	binzad@sysstech.ae	☒	☐	☒	☒
ashishels Dixit	ashishels@multicom.com	☐	☒	☒	☐
TEST1 NETON	shelar@netontechnologies.com	☒	☒	☒	☒

User Account Info

Click on the user name in order to open the User account setting information.

The screenshot displays a web interface for managing users. On the left is a sidebar with navigation options: Dashboard, Numbers, Users (highlighted), Groups, Main Number, Call Logs, Devices, Voicemail Boxes, and Feature Codes. The top header shows 'Total Users 26' and an 'Add User' button. Below the header is a table with columns: Users Settings, Extensions, Phone Numbers, Devices, and User Features. The first row, 'Jim Admin', is highlighted with a red box. To the right of the table, the 'Users Settings' for 'Jim Admin' are shown. A red box highlights the 'User' dropdown menu, which is currently set to 'User'. Other settings include: Name (Jim Admin), Email (jim2995@fonouc.com), VMBox #1021, Send emails to an alternate address (unchecked), Include User in the Company Directory (checked), Ringing Timeout (70 Sec), and Main Extension Number (1021). At the bottom, there are buttons for 'Delete User', 'Cancel', and 'Save Changes'.

This section allows the Administrator to

- Reset user's password
- Reset voicemail box pin
- Change privilege from User to Administrator
- Set user's timezone
- Add to or remove from company directory
- Change extension number

Users Extensions

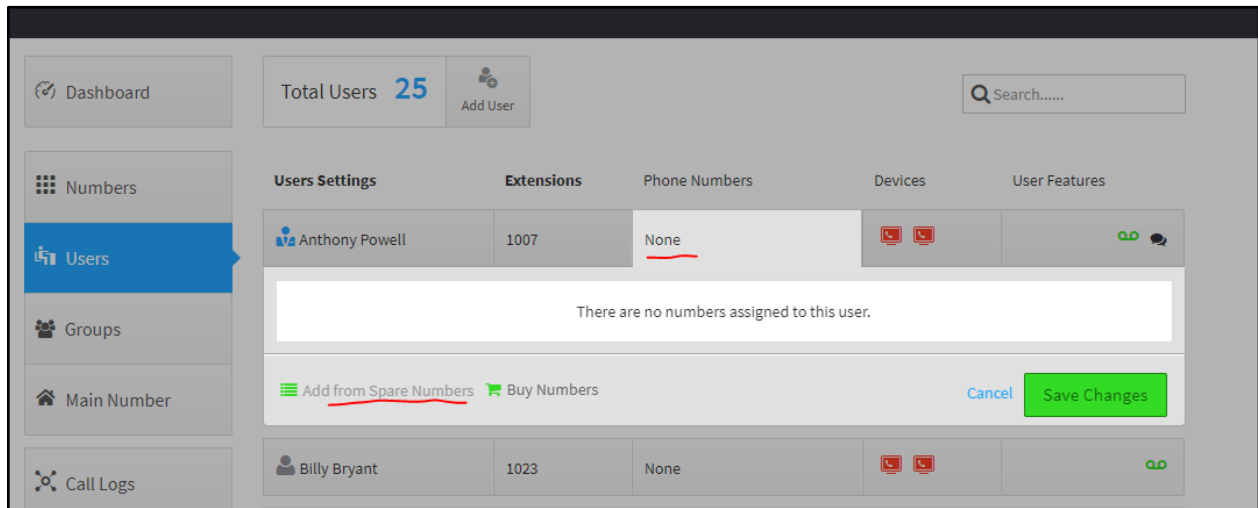
Set the user's extension by clicking on the Extensions tab.
A user can have one or more extensions assigned.

The screenshot shows a web interface for managing users and their extensions. On the left is a sidebar with navigation links: Dashboard, Numbers, Users (highlighted), Groups, Main Number, and Call Logs. The top header displays 'Total Users 26' and an 'Add User' button. A search bar is located in the top right. The main content area has tabs for 'Users Settings', 'Extensions' (selected), 'Phone Numbers', 'Devices', and 'User Features'. Below the 'Extensions' tab, a table lists users and their assigned extensions. The first row shows 'Jim Admin' with extension '1021' (highlighted with a red box), 'None' for phone numbers, and a device icon. Below this row is a form to manage the extension '1021', featuring an 'Un-assign' button and a text input field containing '1021'. Below the form is a section to 'Add an extension to this user' with a 'Cancel' link and a 'Save Changes' button. The second row of the table shows 'Jim1000 Admin' with extension '1000', 'None' for phone numbers, and a device icon.

Users Settings	Extensions	Phone Numbers	Devices	User Features
Jim Admin	1021	None		
Un-assign 1021 + Add an extension to this user Cancel Save Changes				
Jim1000 Admin	1000	None		

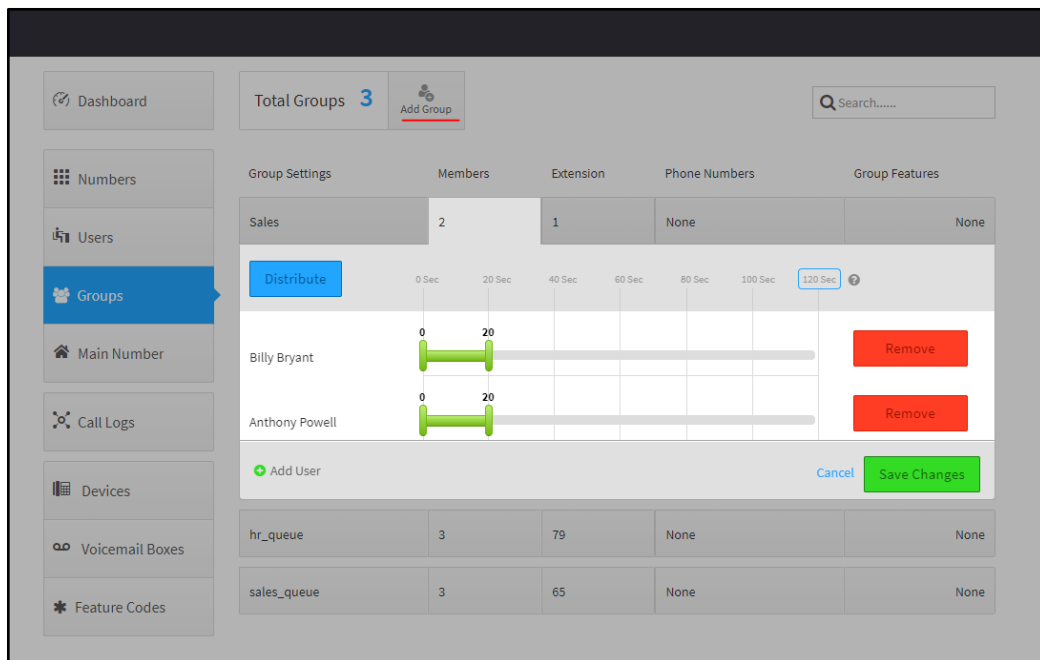
Users Direct Phone Numbers

Set the user's phone by clicking on the Phone Numbers tab. A user can have one or more Phone numbers (DIDs).



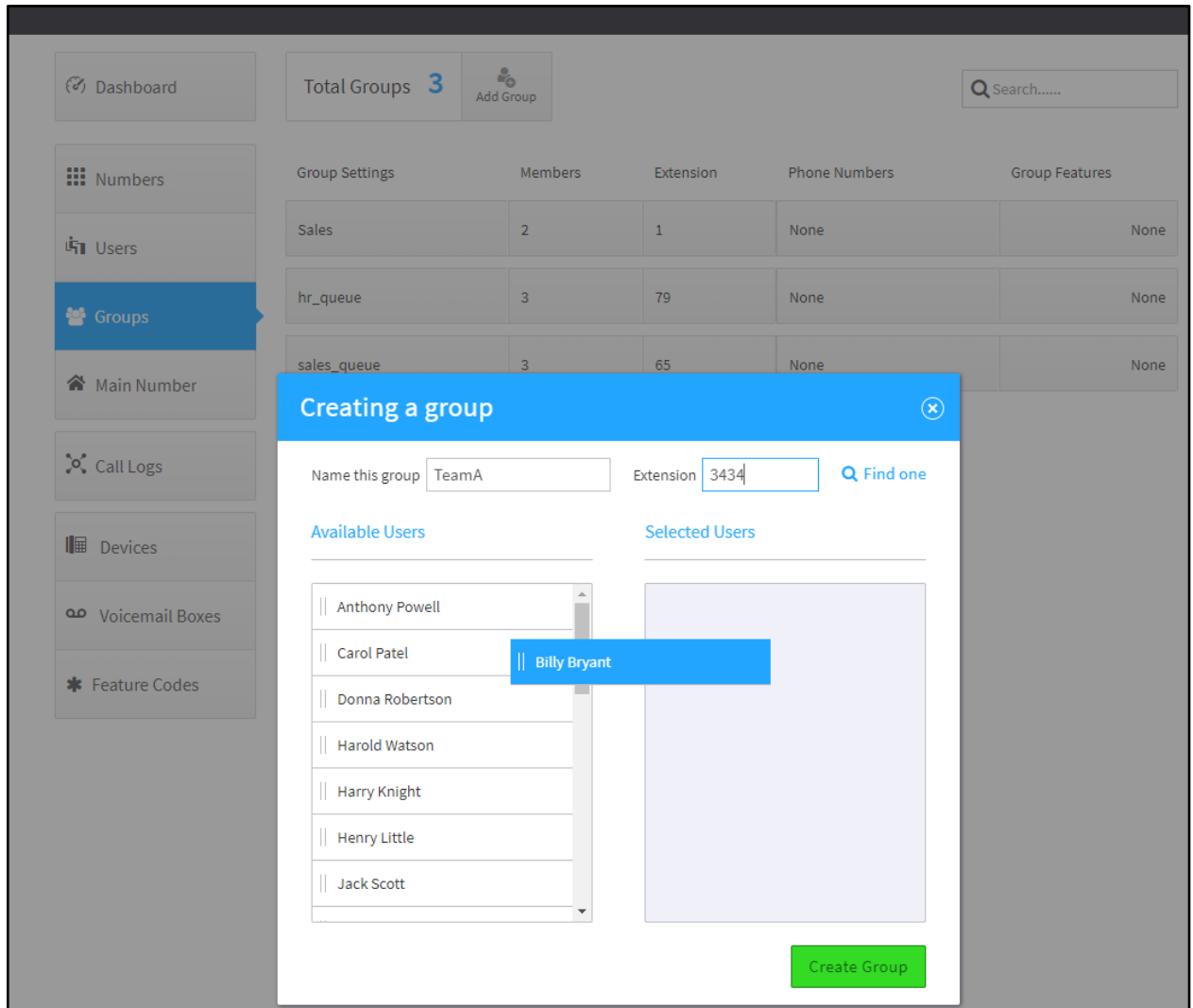
Group Lists

Create a group and add existing users to it. When the group extension is called, all the users will be called and the one that answers will get the call.



1. Name the Group
2. Find an available extension by clicking on Find One or Add one

3. Hold and drop user into Selected user field
4. Create Group



Group Calling Policy

Group ringing can be configured for: Simultaneous, In-Order or Random.

- Simultaneous ring groups will ring all users or devices in a ring group until one answers. This is the DEFAULT strategy when configured from SmartPBX.
- In-Order strategy will call one user/device at a time.
- Random strategy will randomize the users in the group and then dial In-Order

In order to change the default: simultaneous ring strategy in a group, one will have to EDIT a Base Ring group call flow in the Callflow app.

- Navigate to Callflows App

- Click on Callflows
- 1. Search for Base Group call flow for desired Ring Group
- 2. Click on Ring Group in sidebar
- 3. Click to edit the Ring Group
- 4. Change the Ring Strategy
- 5. Save Ring Group

The screenshot displays the 'Callflows' application interface. On the left sidebar, there is a search bar (1) and a list of base groups (2). The main area shows a 'Callflows' tab (3) with a 'Delete callflow' and 'Save changes' button. The 'Ring Group' configuration panel on the right includes fields for 'Name this ring group', 'Ring Strategy' (4), 'Ringback', and 'Number of times to ring members'. A table below shows the 'Ring Group -' configuration with columns for 'Delay (s)', 'Timeout (s)', and 'Delete'. The 'Save' button is labeled (5).

Ring Group -	Delay (s)	Timeout (s)	Delete
sales	0	20	
Sequential	22	20	
Pickup	44	20	

The group can be ringed multiple times before the caller is hung up or passed to the next action.

Group Pickup

When a device that is part of the ring group is ringing in the office but no one is able to pick it up, a group pickup feature will allow others in the group to answer the incoming call by dialing a special extension.

Eg: Incoming call on group Sales (extension 2)

- If nobody is in the office to answer a ringing phone or phones, a user can dial extension 500 in order to answer the call that is currently ringing on a group.

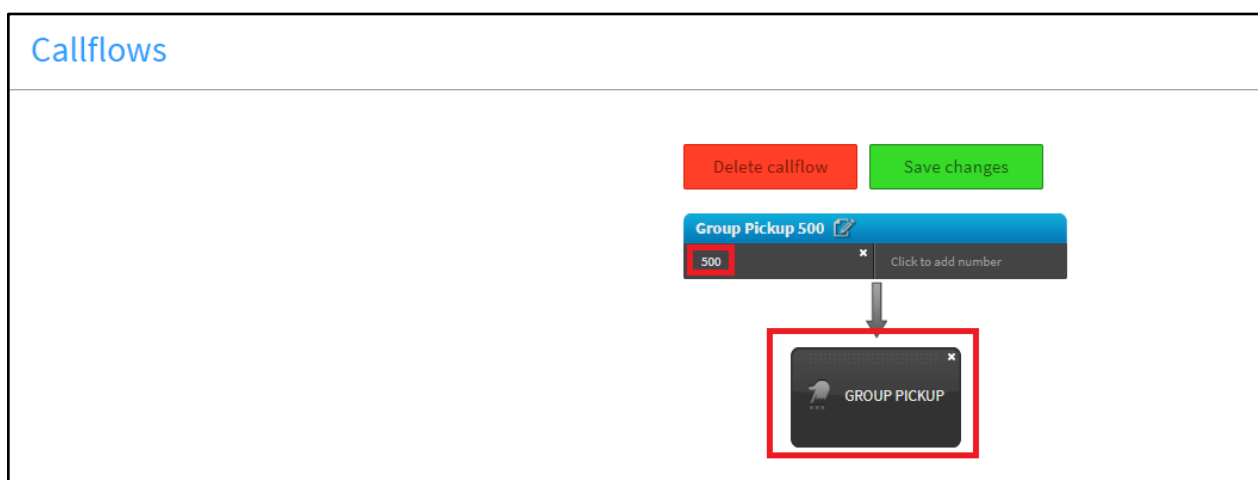
In order to configure the Group Pickup, one must create a Callflow in Callflow app.

- Navigate to Callflows
- Add Call Flow
- Specify an extension, eg: 500
- Drag and drop the “Group Pickup” feature from the left menu
- Double click on “Group Pickup” feature
- Specify a Group that user would be able to pickup from

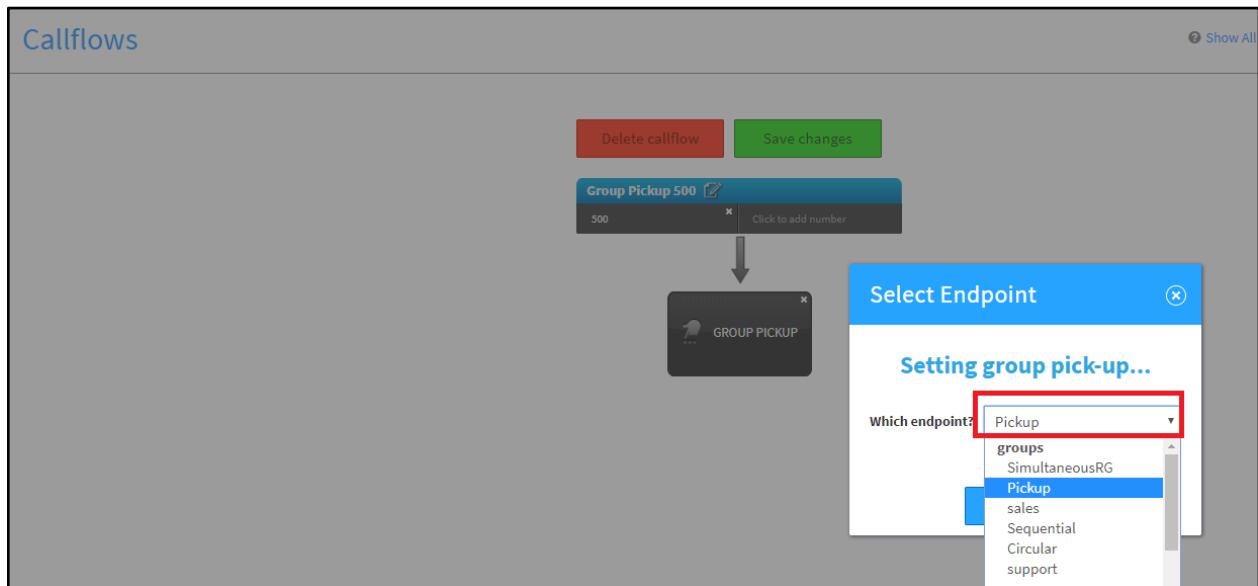
Testing

- From one phone, dial the group that was bound to the group pickup.
- Once the phone is ringing, from another phone dial the above extension 500 in order to pickup the inbound call on call group.

Group Pickup Callflow



Bind group pickup to the group



Devices

Devices are considered user endpoints. Some devices allow sip registrations such as sip phones, mobile sip or desktop clients. Other devices such as cell phones do not have any sip information as they reside on the PSTN and 3G/4G networks.

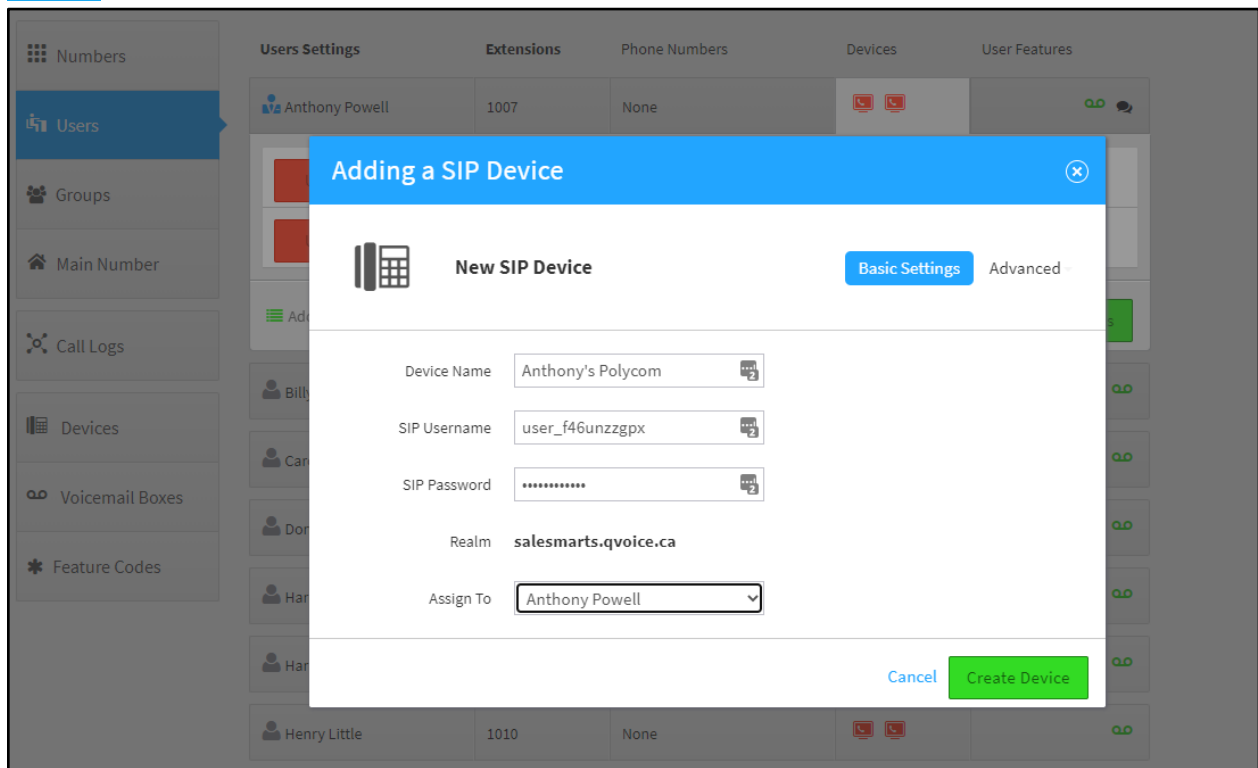
Note: A Device must be bound to a User and a user can have multiple devices associated with it.

Important: Never use the same device credentials on multiple SIP endpoints, this can result in unexpected behaviour as two devices will compete for incoming calls. Always create separate devices for each endpoint. Example: Mobile client and Desktop client.

The screenshot shows a user management interface. On the left is a sidebar with navigation links: Dashboard, Numbers, Users (selected), Groups, Main Number, Call Logs, Devices, Voicemail Boxes, and Feature Codes. The main area has a top bar with 'Total Users 25' and an 'Add User' button. Below this is a tabbed interface with 'Users Settings', 'Extensions', 'Phone Numbers', 'Devices', and 'User Features'. The 'Devices' tab is active, showing a table for user Anthony Powell (ID 1007). The table has columns for user name, extension, phone numbers, devices, and user features. Under the 'Devices' column, there are two rows for devices: 'ucp_mobile' and 'ucp', each with an 'Un-assign' button. Below the table, there are buttons for 'Add from Spare Devices' and 'New Device', along with 'Cancel' and 'Save Changes' buttons. At the bottom, there is a list of other users: Billy Bryant, Carol Patel, Donna Robertson, and Harold Watson, each with their own device assignment status.

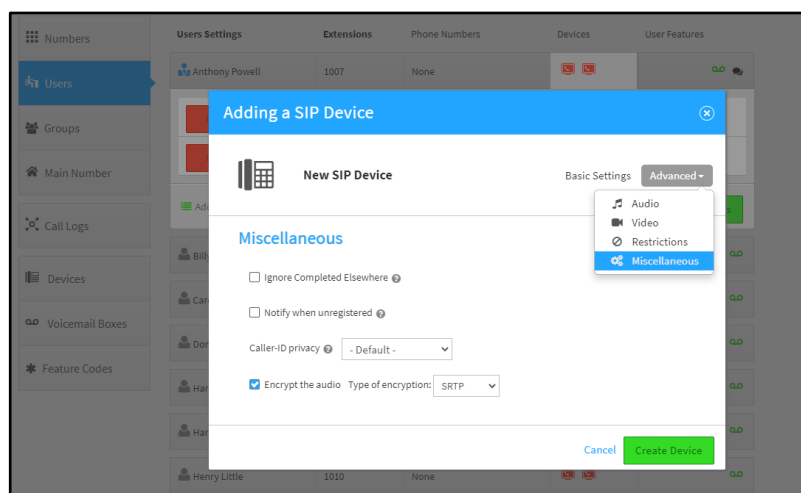
User	Extension	Phone Numbers	Devices	User Features
Anthony Powell	1007	None	Un-assign ucp_mobile Un-assign ucp	
Billy Bryant	1023	None		
Carol Patel	1018	None		
Donna Robertson	1006	None		
Harold Watson	1017	None		

Specify device name: always use a name that associates a device to a user.



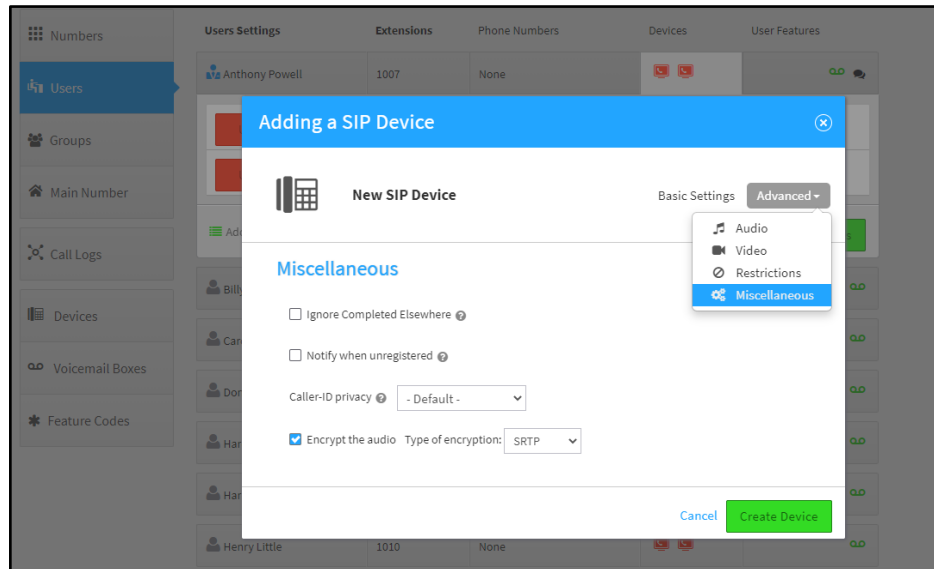
Security: Always let the system specify the random credentials that will be emailed to the end user.

Advanced device configuration: one can specify codecs, secured RTP etc...



Secure RTP

In order to enable secured communications, select the Advanced, Miscellaneous option.



Enable “Encrypt the audio” and specify SRTP. Save configuration and exit the menu.

At this point the device is configured for secured communications. The next step is to configure the endpoint client (softphone or deskphone) that will be registering to this device for TLS/SRTP.

Voicemail Configuration

By default, each created user will get a Voicemail box. When the extension dials the voicemail via *98 the voicemail will prompt the user to set their greeting and set new credentials.

The screenshot shows the 'Voicemail Boxes' management interface. On the left is a sidebar with navigation links: Dashboard, Numbers, Users, Groups, Main Number, Call Logs, Devices, Voicemail Boxes (highlighted), and Feature Codes. The main content area has a header with 'Total Voicemail Boxes 26' and an 'Add Voicemail Box' button. Below this is a table listing existing voicemail boxes.

Number	Name	User	Edit
1007	Anthony Powell's VMBox	Anthony Powell	
1019	Tiffany Washington's VMBox	Tiffany Washington	
1021	Joan Ortega's VMBox	Joan Ortega	
1023	Billy Bryant's VMBox	Billy Bryant	
1024	Jane Hanson's VMBox	Jane Hanson	

The screenshot shows the 'Editing a Voicemail Box' modal for 'Anthony Powell's VMBox'. The modal has a title bar with a close button. Below the title are tabs for 'Basic Settings' (selected) and 'Advanced'. The form contains three input fields: 'Name' (pre-filled with 'Anthony Powell's VMBox'), 'Voicemail Number' (pre-filled with '1007'), and 'PIN Number' (empty). At the bottom, there are three buttons: 'Delete Voicemail Box' (with a trash icon), 'Cancel', and 'Save Changes' (in green).

Editing Greeting - you can basically edit each users voicemail box here or from the user screen as mentioned before. You have access to several Options, Recipients, Greeting Medias

The screenshot displays a web interface for managing voicemail boxes. On the left is a sidebar with navigation links: Dashboard, Numbers, Users, Groups, Main Number, Call Logs, Devices, Voicemail Boxes (highlighted), and Feature Codes. The top header shows 'Total Voicemail Boxes 26' and an 'Add Voicemail Box' button. A table lists voicemail boxes with columns for Number, Name, User, and Edit. The first entry is '1007' for 'Anthony Powell's VMBox' by 'Anthony Powell'. An 'Editing a Voicemail Box: Anthony Powell's VMBox' modal is open, showing settings for 'Anthony Powell's VMBox'. It has tabs for 'Basic Settings' and 'Advanced'. The 'Options' section includes dropdowns for 'Timezone' (set to '- Account/User default -') and 'Media Extension' (set to 'mp3'). Below these are checkboxes for 'Already Setup?', 'Require PIN?', 'Auto-Login Enabled?' (checked), 'Skip Greetings?', 'Skip Instructions?', 'Delete after Notification?', and 'Configurable via Menu?' (checked). At the bottom of the modal are buttons for 'Delete Voicemail Box', 'Cancel', and 'Save Changes'.

Number	Name	User	Edit
1007	Anthony Powell's VMBox	Anthony Powell	

Editing a Voicemail Box: Anthony Powell's VMBox

Anthony Powell's VMBox

Basic Settings | **Advanced**

Options

Timezone: - Account/User default -

Media Extension: mp3

☐ Already Setup?

☐ Require PIN?

☒ Auto-Login Enabled?

☐ Skip Greetings?

☐ Skip Instructions?

☐ Delete after Notification?

☒ Configurable via Menu?

Delete Voicemail Box

Cancel | Save Changes

Options: Set Timezone for remote users, Media type (mp3 or wav)

Recipients - add additional recipients of the voicemail box

The screenshot shows a web interface for editing a voicemail box. The title bar is blue and says "Editing a Voicemail Box: Anthony Powell's VMBox" with a close button. Below the title bar, there's a header area with a logo, the name "Anthony Powell's VMBox", and tabs for "Basic Settings" and "Advanced". The "Advanced" tab is selected. The main section is titled "Recipients" in blue. Below the title, there's an information icon and text: "This panel lets you manage who will be notified when someone leaves a message in this voicemail box. Click on 'Add New Recipient' to add an e-mail address to the list". There's a dashed box containing a text input field with "new@fonouc.com", an "Add" button, and a "Cancel" button. At the bottom, there's a "Delete Voicemail Box" button with a trash icon, a "Cancel" button, and a "Save Changes" button.

Greeting Media - option to upload a greeting, click on Upload and browse for greeting(s). Conversely every user has the ability to record their voicemail using Feature code *97

The screenshot shows the same web interface as the previous one, but the "Greeting Media" section is active. The title bar and header are the same. The "Greeting Media" section has a blue title. Below the title, there are two rows. The first row has a "Media:" label, a dropdown menu showing "- None -", and an "Upload" button. The second row has a "Temporary:" label, a dropdown menu showing "- None -", and an "Upload" button. At the bottom, there's a "Delete Voicemail Box" button with a trash icon, a "Cancel" button, and a "Save Changes" button.

Feature Codes

Feature codes can be configured in Callflows application.

The screenshot displays the 'Available Feature Codes' section within the Callflows application. On the left is a sidebar menu with options: Dashboard, Numbers, Users, Groups, Main Number, Call Logs, Devices, Voicemail Boxes, and Feature Codes (highlighted in blue). The main content area is titled 'Available Feature Codes' and lists various feature codes categorized into Call-Forward, Hot-Desking, Miscellaneous, and Parking. Each code includes a help icon (question mark) and a code value.

Category	Feature Code	Code Value
Call-Forward	Update Call-Forward	*56
	Enable Call-Forward	*72
	Disable Call-Forward	*73
	Toggle Call-Forward	*74
Hot-Desking	Enable Hot Desking	*11
	Disable Hot Desking	*12
	Toggle Hot Desking	*13
Miscellaneous	Direct to Voicemail	**
	Intercom	*0
	Move	6683
	Privacy	*67
	Check Voicemail	*97
	Check Voicemail (auto login)	*98
Parking	Park and Retrieve	*3
	Valet	*4
	Retrieve	*5

Call Logs

You can check log files for all incoming and outgoing calls made through Multycomm PBX. You can search for numbers, sort, search by date range. These log files are also downloadable

Call logs can be obtained via SmartPBX or Portal. Portal provides much more extensive CDR and call reports including user analytics.

Call Logs

Today This Week This Month Custom

Download ?

	Date/Time GMT-05:00	From	To	Duration (Min)	Hangup Cause	Actions
←	07/01/21 21:45:05	1007 Anthony Powell	+1 905 444 5557 19054445557	0:04	normal_clearing	⬆
←	07/01/21 21:44:50	1007 Anthony Powell	+1 905 444 5556 19054445556	0:05	normal_clearing	⬆
←	07/01/21 21:44:36	1007 Anthony Powell	+1 905 444 5555 19054445555	0:04	normal_clearing	⬆

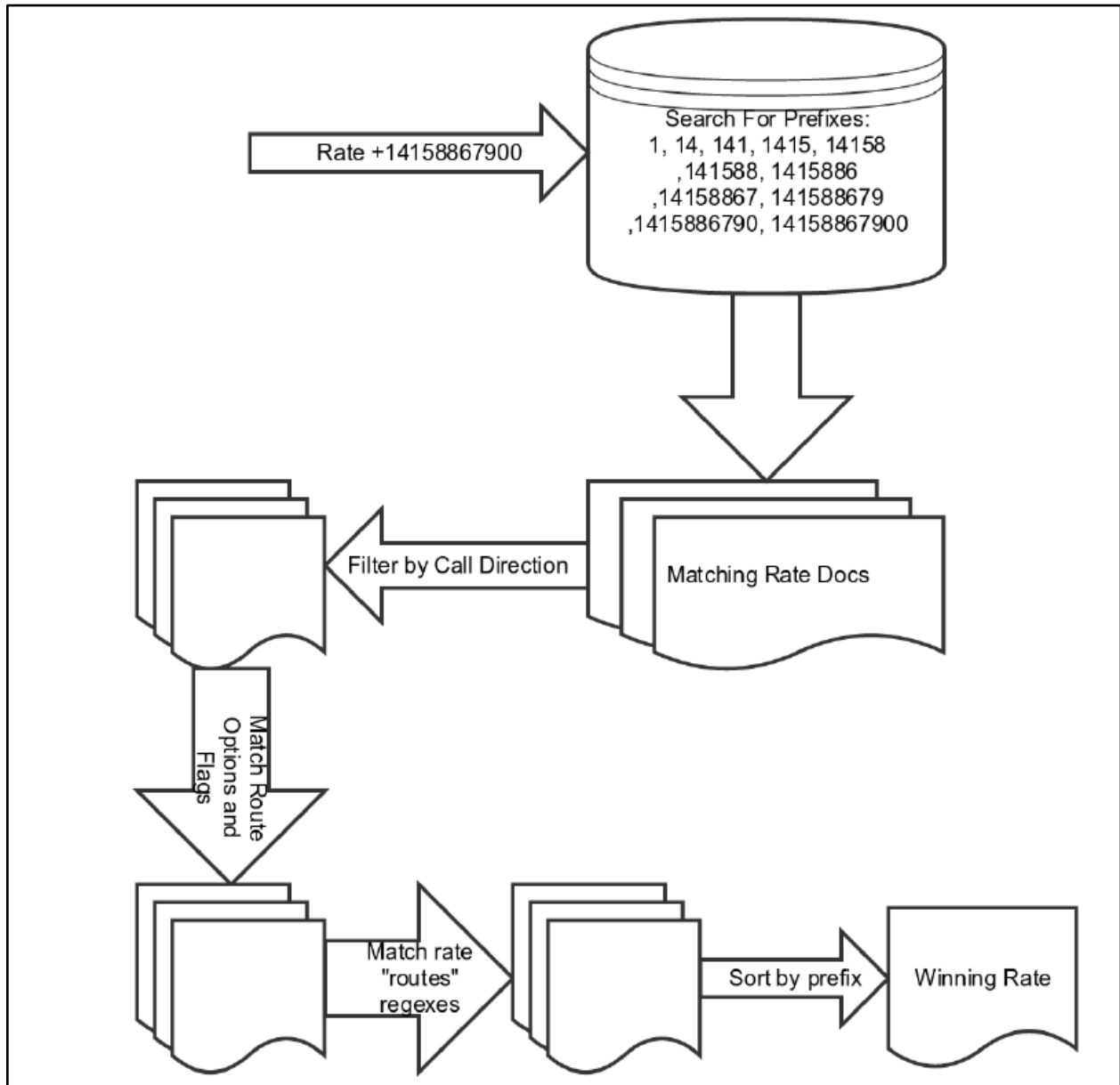
Billing

The Multycomm platform supports following billing features:

- Recurring Subscriptions (monthly charges)
- Pro-rated monthly subscriptions
- Activation charges
- Per-minute / per-use charges
- Limits on spend (daily, monthly, pre-pay/credit base)
- Allocations of usage (400 minutes included - daily, monthly, etc.)
- Bursting / overage charges
- Rating of calls
- Prepay vs. postpay
- Auto top up
- Deposit tracking
- Accounts Payable / Accounts Receivable

During account creation, the account is assigned a Service Plan that contains a global or per-account per-minute rate deck as well as per month subscription. Both the rate deck and the monthly subscription information can be customized per account.

- Strategies for warning customers
 - Low balances
 - Overdue
- Discounts
 - One time
 - Continuous
- Rate decks (global + account based)
- Actual cost analysis



Service Plan

Service Plan is used to define how and what the customer should be billed for. Almost every feature of the platform can be “charged for”: the end customer will be prompted to accept new charges when adding new features or creating resources such as users and devices.

Service plan can be overridden, admin can choose to provide special per user, DID or porting pricing or offer a discount from the MSRP. Furthermore, Admin can set up multiple Service Plans that are tied to custom Rate Decks that are used for per-minute call rating and billing.

Service Plan Administration

The Service plan can be created in the Administration Portal: <https://<domain>/portal>.

Account

Phone System

Storage

Numbers

Recordings

Billing

Rate decks

Service Plans

Payment Gateway

Service Plans

+ Add

ID	Name	Ratedeck	Actions	
gcc_default_sp	GCC Default Service Plan	ratedeck_ps		

Rate Deck

System operators can create multiple rate decks and assign those ratedecks to their accounts or sub-accounts via service plans.

rate_cost	description	name	iso_country_code	prefix	routes	ratedeck_id
0.1	UK	UK	UK		44 ["^\\+?44(\\d*)\$"]	ratedeck_gold
0.1	RS	RS	RS		38 ["^\\+?38(\\d*)\$"]	ratedeck_gold
0.02	US	US	US		1 ["^\\+?1(\\d*)\$"]	ratedeck_gold
0.25	US	US	US		1 ["^\\+?1208(\\d*)\$"]	ratedeck_gold
0.3	UK	UK	UK		44 ["^\\+?44(\\d*)\$"]	ratedeck_silver
0.4	RS	RS	RS		38 ["^\\+?38(\\d*)\$"]	ratedeck_silver
0.52	US	US	US		1 ["^\\+?1(\\d*)\$"]	ratedeck_silver
0.65	US	US	US		1 ["^\\+?1208(\\d*)\$"]	ratedeck_silver

Rate Deck Administration

The Rate decks can be created in the Administration Portal: <https://<domain>/portal>.

Account

Phone System

Storage

Numbers

Recordings

Billing

Integrations

CRM

Ratedecks

2 Ratedecks

ratedeck 11 Rates

Rate Number

View Only

Download

Add Rate

Filters

Name

Description

Prefix

Cost

Routes

Name	Description	Prefix	Cost	Rate Increment	Routes	Actions
DFLT	DFLT	0	0.2	60	^\\+?70,+\$	 
DFLT	DFLT	2	0.2	60	^\\+?72,+\$	 
DFLT	DFLT	3	0.2	60	^\\+?73,+\$	 
DFLT	DFLT	4	0.2	60	^\\+?74,+\$	 
DFLT	DFLT	5	0.2	60	^\\+?75,+\$	 

v1.95.12

Billing Overview

The Admin can access the billing reports as part of the Account View. The master account has the global view and can see each transaction for each multi-tenant account.

IM Reports

Provision

Status

Config

Users

SmartPBX

CallCenter

Account Admin (admin)

salesmarts child of admin

Usage Charges

Total Calls3

Total Minutes3

Total Amount0.60

Download Usage Charges

Start Date

06/30/2021 12:00 AM

End Date

07/02/2021 12:17 AM

Timezone

America/New_York

Extension

All

Get Usage Charges

Filters

All fields

Extension

Start Time

Caller ID Name

Caller ID Number

Show More Filters

Oldest Record Fetched: 2021-07-01 21:44:40 - Newest Record Fetched: 2021-07-01 21:45:09

Start Time	Extension	Caller ID Name	Caller ID Number	Direction	Callee ID Name	Callee ID Number	Duration	Rate	Amount
2021-07-01 21:45:09	1007	Anthony Powell	1007	outbound	19054445557	19054445557	1 (min)	0.2	0.20
2021-07-01 21:44:55	1007	Anthony Powell	1007	outbound	19054445556	19054445556	1 (min)	0.2	0.20
2021-07-01 21:44:40	1007	Anthony Powell	1007	outbound	19054445555	19054445555	1 (min)	0.2	0.20

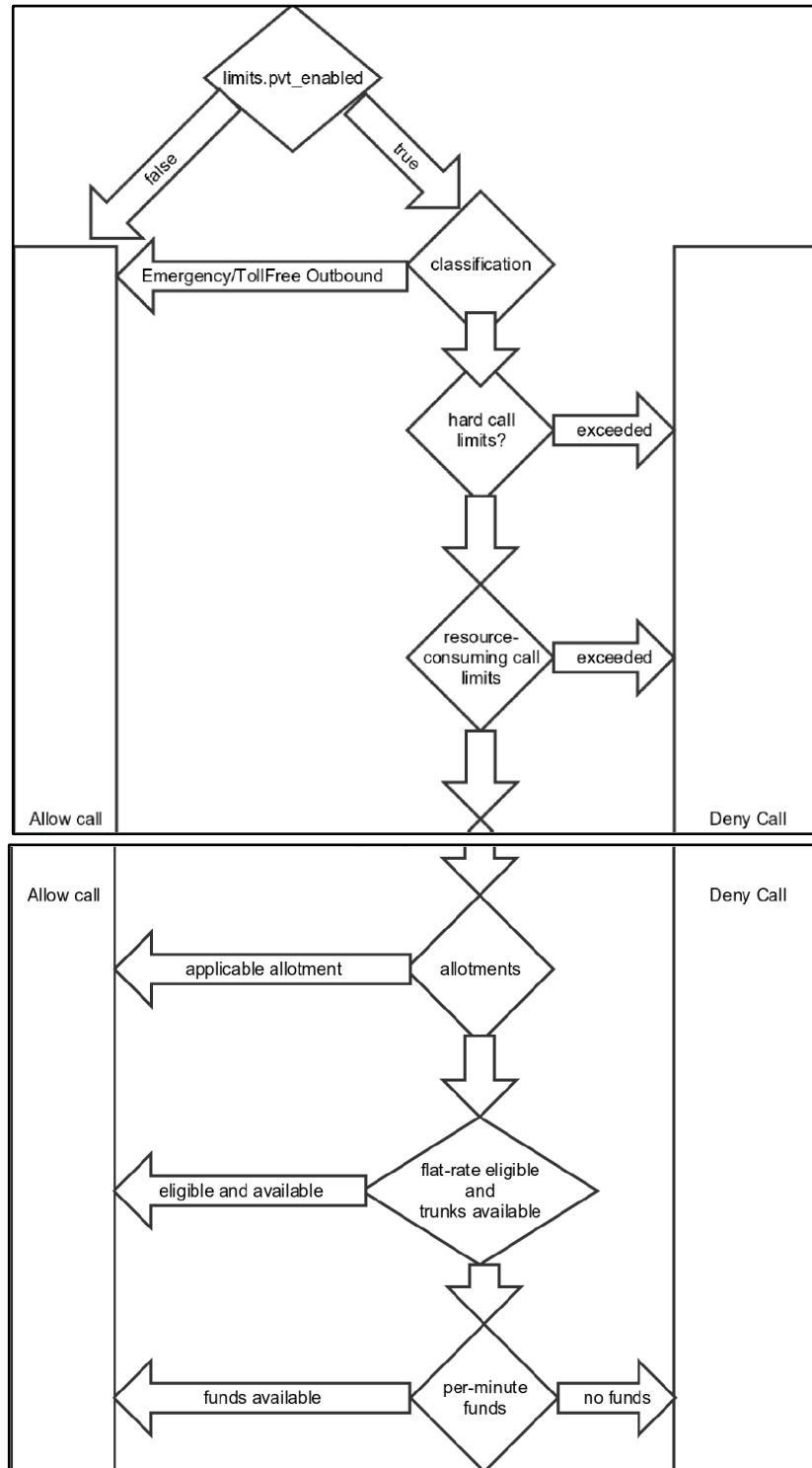
Limits

- Inbound
 - Limit the number of simultaneous inbound calls that can be received
- Outbound
 - Limit the number of simultaneous outbound calls that can be made
- Two-way
 - Limit the number of simultaneous calls that can be made
- Resource consuming
 - Any endpoint the system operators likely pay for (upstream carriers generally)
 - Limit the number of calls that can consume resources (internal calls unaffected)

The screenshot displays the 'Accounts' management interface. At the top, there's a search bar with 'demo' and a 'Filter by Account Status' dropdown. Below this is a table with columns: Name, Domain, Users, Devices, Status, Credit, Call Limits, and Service Plan. The table lists several accounts, including 'ambientdemo', 'bulldemo', 'bzdemo', 'demo1', 'demo123', 'demo2', 'demo33', 'demo4', 'demo5', and 'demo6'. A modal window titled 'Simultaneous Calls Limits' is open, allowing configuration for 'Inbound Trunks' (set to 0), 'Outbound Trunks' (set to 0), and 'Two-way Trunks' (set to 2). The modal has 'Cancel' and 'Apply' buttons. A red arrow points to the 'Call Limits' column in the table, specifically to the value '2' for the 'ambientdemo' account.

Name	Domain	Users	Devices	Status	Credit	Call Limits	Service Plan
ambientdemo	ambient.qvoice.net	2	5	Active	100	2	service_plan_default
bulldemo	bulldemo.qvoice.net	3	4	Active	100	10	service_plan_default
bzdemo	bzdemo.qvoice.ca	4	6	Active	96	10	service_plan_default
demo1	demo1.qvoice.net	3				10	service_plan_default
demo123	demo123.qvoice.ca	0				10	service_plan_default
demo2	demo2.qvoice.net	0				2	service_plan_default
demo33	3bd6e4.qv1.fonouc.com	0				2	service_plan_default
demo4	demo4.qvoice.net	0				15	service_plan_default
demo5	demo5.qvoice.net	0				5	service_plan_default
demo6	demo6.qvoice.net	0	0	Active	100	10	service_plan_default

- Burst
 - Allows account to consume more trunks than the base number allotted, typically for short intervals
 - Good for seasonal, customer support, radio shows, call centers, schools, etc
- Bundled Trunks
 - Inbound, Outbound, Two Way as well
 - Limit determined by the number of users or devices (configurable)
 - "twoway_bundled":"user", for example
- Allotments
 - Buckets of minutes per time-period Monthly, Weekly, Daily, Hourly, Minutely (seriously)
- Emergency calls are immediately authorized, as are outbound calls to toll free numbers.
- Each CDR is augmented with two fields showing the trunk usage for the account and the reseller at the time of the call.



Trunk/Carrier Configuration

SIP Trunks, also called Carriers, connect the Multycomm platform to the carrier PSTN network. They facilitate outbound calls from Multycomm platform to the PSTN, as well as, inbound calls from the PSTN.

SIP Trunks can be created for each account separately. However, in order to avoid duplicating sip trunk configuration for each account, admin can configure the account to use the sip trunk defined by its parent account.

1. Log into Multycomm portal using super admin account credentials
2. Select Admin account from the top right
If you used super admin credentials the account will say admin
3. Select Config -> Trunks from the menu
Configure Route to Trunks

Name	Enabled	Emergency	Weight cost	DID Routing	Actions
GH4	true	false	50	^\+([1\d{6,})\$	
GH1	true	false	50	^\+([1\d{6,})\$	
GH3	true	false	50	^\+([1\d{6,})\$	
GH2	true	false	50	^\+([1\d{6,})\$	
PBXACT_Trunk	true	false	50	^\d{3}\$	

4. Route to trunks:
Options are: **Parent** or **Local**
 - a. **Parent**
Use trunks configured on the parent account.
In this mode, no local trunks need to be configured, as the parent account sip trunks will be used by this account.

Since Admin account is the top most account, its Route should be Local

b. **Local**

Use trunks configured for this account

5. Create Local SIP Trunk

 + Add

Click on to create a new trunk

The sip trunk configuration is currently in JSON format in order to facilitate flexibility.

By default the sample json config file will be provided so the user can just fill in the blanks.



```
1 {
2   "name": "Trunk1",
3   "enabled": true,
4   "emergency": false,
5   "flags": [
6     "fax"
7   ],
8   "gateways": [
9     {
10      "server": "call4567@airtel.com",
11      "channel_selection": "ascending",
12      "codecs": [
13        "PCMU",
14        "PCMA"
15      ],
16      "media": {
17        "fax_option": true
18      },
19      "enabled": true,
20      "endpoint_type": "sip",
21      "force_port": false,
22      "invite_format": "route",
23      "password": "",
24      "port": 5060,
25      "prefix": "",
26      "progress_timeout": 20,
27      "realm": "",
28      "skype_r": false,
29      "username": ""
30    }
31  ]
32 }
```

6. Mandatory Fields

a. **Name**

Descriptive name of the sip trunk

b. **Realm**

SIP Trunk domain name, or an IP address

c. **Server**

SIP Trunk IP address

d. **Prefix**

Used to add a digit in front of the outbound dialed number

e. **Gateways**

A SIP trunk can contain one or more gateways.

This allows one to create a load balanced sip trunk to two carrier servers.



Calls will be routed based on the gateway weights.

Refer to two gateway sip trunk in example below.

f. **Rules**

Used to define which outbound dial rules.

If the outbound did/number matches the rule of this sip trunk, then the matched sip trunk will be used to send the outbound calls.

The match must be performed using the E164 number format.

"^\\+1(\\d{10})\$" : Match US numbers and remove +1

"^\\+(1\\d{10})\$" : Match US numbers and remove + only

7. Save Configuration

Custom Routing

Account custom routing enables two features

- Access PIN per user
 - User must access a PIN in order to be granted the ability to dial out
 - PINs can be used to restrict International or high rate calls
- Custom Caller ID
 - Ability to set custom caller ID per user

Access Pin per User

Navigate to the Administration portal <https://<domain>/portal>

- Select Config -> Navigate to Pins.
- Enable Pin Feature
- Create a Pattern Group
- Bind a Pattern Group to a PIN

The screenshot displays the 'Pins' configuration interface. On the left is a sidebar with navigation options: Account, Phone System, Trunks, Pins, Caller ID, Locations, Queues, Faxboxes, and Storage. The main content area is titled 'Pins' and includes a 'Pins Feature' toggle (currently 'Enable') and a 'Pins Mode' dropdown (currently 'Whitelisted'). Below these are five tabs: 'PATTERNS GROUPS', 'PINS GROUPS', 'ENABLE PINS PER USER', 'IGNORE PINS', and 'IGNORE PINS ON TRUNKS'. The 'PATTERNS GROUPS' tab is selected, showing a '+ Add Group' button and a table. The table has columns for 'Group Name', 'Patterns', and 'Actions'. One group is listed: 'STD' with the pattern '**9876#' and edit/delete icons.

Feature	Description
Whitelist	Enable PIN feature for ALL users. Admin has the ability to “whitelist” specific users that are allowed to bypass the pin. Otherwise all other users must specify the pin before dialing external numbers.
Blacklist	Only enable PIN for specific “blacklisted” uses. Admin has the ability to “blacklist”/enable pin per user.

Ignore PIN on trunks

- Specify a trunk that would allow users to dial without being asked for a PIN
- This is useful, if a trunk is defined to only dial low rate or free numbers.

Pins

Pins Feature

☐ Enable

Pins Mode

Whitelisted

PATTERNS GROUPS	PINS GROUPS	ENABLE PINS PER USER	IGNORE PINS	<u>IGNORE PINS ON TRUNKS</u>
PINS will be ignored if call is routed to selected trunks				
	Name			Ignore Pins
	AWS Voice Connector			☐
	ThinQ Voice			☐
	startelecom			☐

Pin Patterns and Assignments

The combination of PINs and Groups allow administrators to set up complex outbound routing rules that would restrict high rate or international calls per user.

Feature	Description
Group	Used to define one or more DID/Phone Number patterns. The outbound call number must match a group pattern before it will be routed.
PIN	A PIN is a numeric code that is mapped to one or more Groups. Outbound caller must dial a correct PIN before the call will be routed. The PIN also allows the admin to specify a Note or a Tag to map the PIN to a user or a user accounting number.

In this example, the Group **Local** will only route calls starting with 416 or 647 numbers. Adam's pin is 1242314 and he will only be able to dial to 416 and 647 numbers.

Patterns group

Group Name

Local

Patterns

416
647

Save

Assign groups to pin

Pin

1242314

Note

Adam Michaels in Accounting

Groups

Local

Save

Depending on “whitelist” or “blacklist” mode then user selection will perform different function

- In whitelist mode

- Each selected user will be allowed to bypass PIN
- Furthermore,
- In blacklist mode
 - Each selected user will be forced to enter a PIN, all other users will not need to enter the PIN
 - Furthermore, if caller id was set, the outbound caller id would be overwritten to the custom set value.
 - In example below, Callerid Name: Anthony Callerid Number: 1007 will be changed to Tony 4165556666

Custom CallerID

- if caller id was set, the outbound caller id would be overwritten to the custom set value.

Account
Phone System
Trunks
Pins
Caller ID
Locations
Queues
Faxboxes
Storage
Numbers

Custom Caller ID

Custom Caller ID Feature
☐ Enable
 Save

Custom Caller ID Filters

Username	Name	Extension	Custom Caller ID Name	Custom Caller ID Number
abhidixit.dcs@gmail.com	ABHISHEK DIXIT	404	Caller ID Name	Caller ID Number
thomas@fonouc.com	Thomas Ripari	2255	Caller ID Name	Caller ID Number
vipul.bharadwaj22@gmail.com	vipul bharadwaj	103	Caller ID Name	Caller ID Number
rizowan@ssd-tech.io	Rizowan bakr	201	Caller ID Name	Caller ID Number

Call Recording

Administrators can enable call recording per Account or per User. When enabling call recording it is recommended that only External calls are recorded in order to save on disk space.

Account

Phone System

Storage

Numbers

Recordings

Call Recording

Billing

Integrations

Call Recording

Save Config

☒ ENABLED Mode

ACCOUNT

 Total Storage Size Limit (MB)

0

 Retention Duration (Days)

90

Custom Filename

Account Configuration

☒ Inbound Internal ☐ Outbound Internal

☒ Inbound External ☒ Outbound External

Example of per user call recording.

Call Recording

Save Config

☒ ENABLED Mode

PER USER

 Total Storage Size Limit (MB)

0

 Retention Duration (Days)

90

Custom Filename

Users Configuration

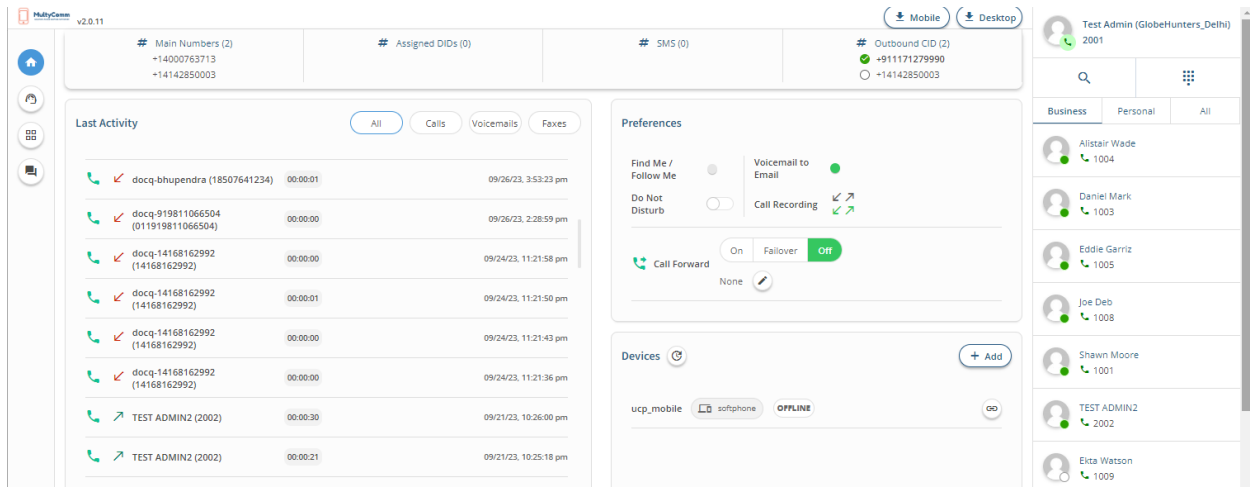
Filter by Name or Username

☐ Inbound Internal ☐ Inbound External ☐ Outbound Internal ☐ Outbound External

Name	Username	Inbound Internal	Inbound External	Outbound Internal	Outbound External
Shahab NETON2	support@netontechnologies.com	☐	☒	☐	☒
Aweek Roy	aveek@mutycomim.com	☒	☒	☐	☒
Ameer Lithe	ameer@lithesd.com	☐	☒	☒	☒
TEST1 NETON	shelar@netontechnologies.com	☒	☐	☒	☐

User Control Panel

User control panel offers the extension owner/user full extension configuration control. It further provides integrated call history as well as Desktop voice calling directly from within the browser via WebRTC. UCP is located at ***https://<cloud pbx domain>/ucp***



UCP Device

In order to enable WebRTC calling within the UCP, the Admin must create a “ucp” named device in SmartPBX.

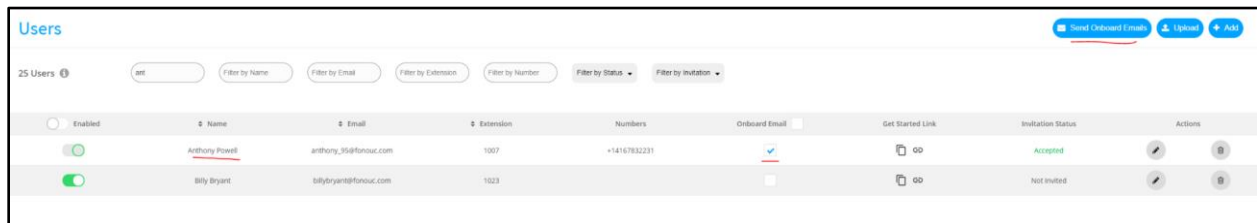
Device Name: ucp

Device Type: Soft Phone

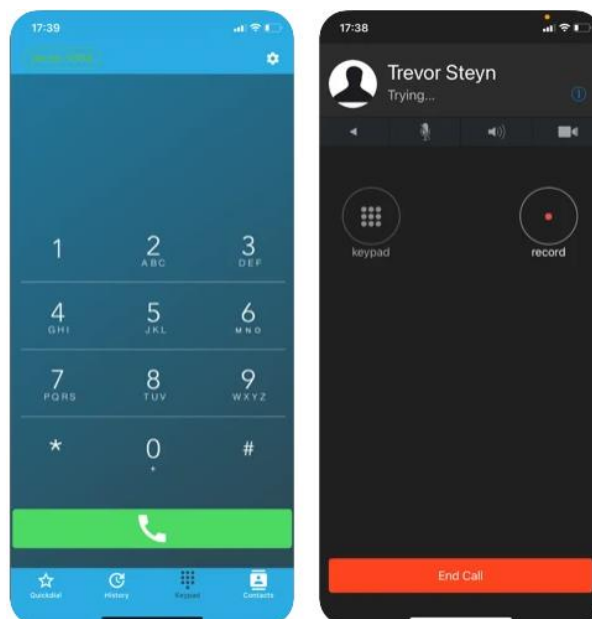
UCP URL: ***https://<cloud pbx link>/ucp***

Mobile App

Mobile users are onboarded via Portal Invitation. Users will receive a link to the app and a login magic link.



Users								Send Onboard Email	Upload	Add
25 Users ⓘ										
Enabled	Name	Email	Extension	Numbers	Onboard Email	Get Started Link	Invitation Status	Actions		
☐	Anthony Powell	anthony_35@fonuc.com	1007	+14167832231	☒	GO	Accepted	Edit	Delete	
☒	Billy Bryant	billybryant@fonuc.com	1023		☐	GO	Not invited	Edit	Delete	

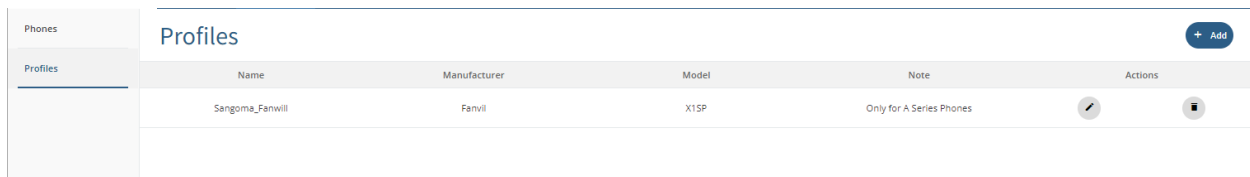


IP Phone Provisioning

In this section we will outline all the steps necessary to configure endpoint provisioning and how to add endpoints to be provisioned.

Summary

1. Log into the Portal with the admin user of the account you're looking to add endpoints
 - a. **https://<domain>/portal**
 - b. User: your email
 - c. Pass: <password>
 - d. Account: Name of account without "space" or "-"
2. Select **Provisioning** on the top navigation bar



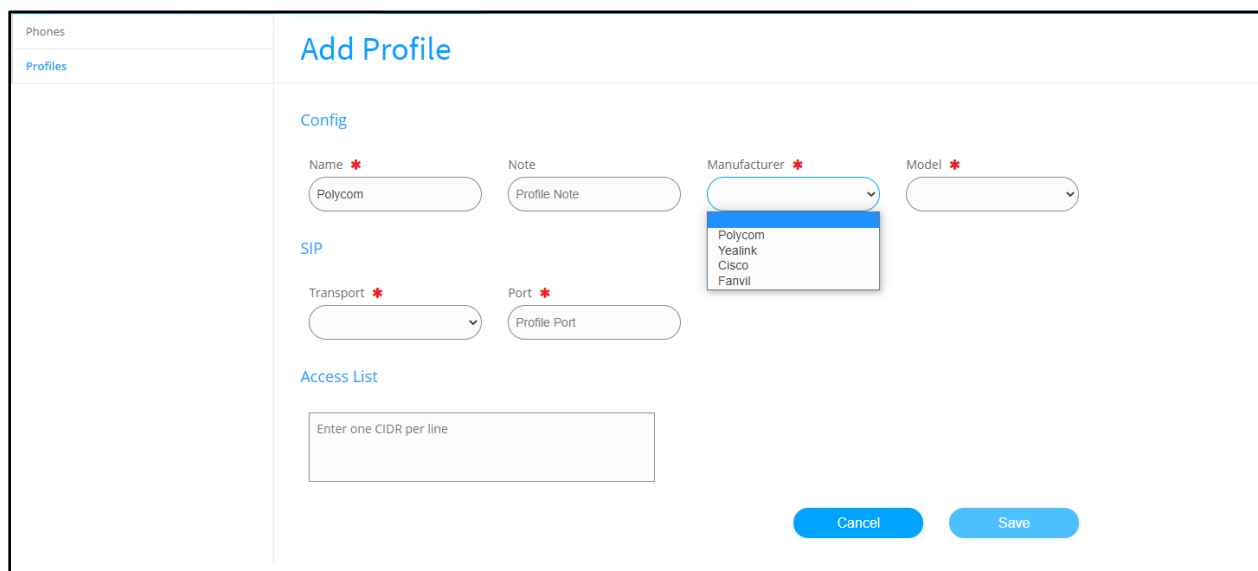
The screenshot shows a web interface with a left sidebar containing 'Phones' and 'Profiles' (selected). The main area is titled 'Profiles' and contains a table with the following data:

Name	Manufacturer	Model	Note	Actions
Sangoma_Fanwill	Fanwill	X1SP	Only for A Series Phones	 

An '+ Add' button is located in the top right corner of the table area.

3. Select Profiles on the *left* navigation bar

4. Click *Add* to add a new profile



The screenshot shows the 'Add Profile' form with the following sections:

- Config**
 - Name * (text input: Polycom)
 - Note (text input: Profile Note)
 - Manufacturer * (dropdown menu: Polycom, Yealink, Cisco, Fanwill)
 - Model * (dropdown menu)
- SIP**
 - Transport * (dropdown menu)
 - Port * (text input: Profile Port)
- Access List**
 - Text area: Enter one CIDR per line

Buttons: Cancel, Save

- a. Fill in the form for the desired model and make of the endpoint
 - b. Save the form, you'll be redirected back to the *Profiles* page
5. Click on the *Phones* page on the left navigation bar

6. Click the add button to *Add* a phone
 - a. Fill in the form with the MAC Address of the endpoint you're looking to provision
 - b. Select the profile associated with that phone
 - c. Configure its lines and buttons
 - d. Save the form, you'll be redirected back to the *Phones* page
7. Change the endpoint configuration to use the provisioner at: **https://<your-domain>:9443/provisioning/<manufacturer>**
 - a. <your-domain> refers to the the FQDN in which your system is hosted
 - b. <manufacturer> is the make of the endpoint. Currently supported makes are *polycom* and *yealink*
8. Yealink Example:
 - a. Open Yealink WebGUI
 - b. In the Auto Provisioning section of the UI add
 - i. https://<your-domain>:9443/provisioning/yealink
 - ii. Save
9. Polycom Example:
 - a. Open Polycom WebGUI
 - b. In the Auto Provisioning section of the UI add
 - i. https://<your-domain>:9443/provisioning/polycom
 - ii. Save
10. Fanvil Example:
 - a. Open Fanvil WebGUI
 - b. In the Auto Provisioning section of the UI add
 - i. https://<your-domain>:9443/provisioning/fanvil
 - ii. Save

Phone Profiles

Profiles are common settings to be applied to a group of endpoints of the same make and model. These settings are specific for each account and cannot be shared amongst multiple accounts.

Currently, the only setting available is the wallpaper on select models of yealink but more settings are becoming available soon as we expand our selection of provisionable endpoints

Phones

Profiles

Add Profile

Config

Name *

Polycom

Note

Profile Note

Manufacturer *

Polycom

Yealink

Cisco

Fanvil

Model *

SIP

Transport *

Port *

Profile Port

Access List

Enter one CIDR per line

Cancel

Save

NWAY Calling

During the active call, Jim can dial a DTMF feature code to trigger a Blind or Attended Transfer to another user. The Blind transfer allows Jim to connect to connect Bob who he is currently talking to, with Alice, using *2<extension>* without waiting for Alice to answer. The Attended transfer allows Jim to put Bob on hold and call Alice, using *4<extension>*, when Alice answers, Jim can initiate the transfer via hangup and connect Bob and Alice together. Alternatively Jim can dial *3 and put Bob, Alice and Jim into a conference. From within the conference, any member can initiate another Attended Transfer via *4<extension>* and bring in another user to the conference.

Blind Transfer		
Person who is in the PBX	*2<extension>*	Make a blind transfer
Attend Transfer		
Person who is in the PBX	*4<extension>*	Make an attended transfer
Person who is in the PBX (Caller)	*3	Add newly connected party to the conference
Person who is in the PBX (Caller)	*1	Drop called party and connect to other(s) in the call

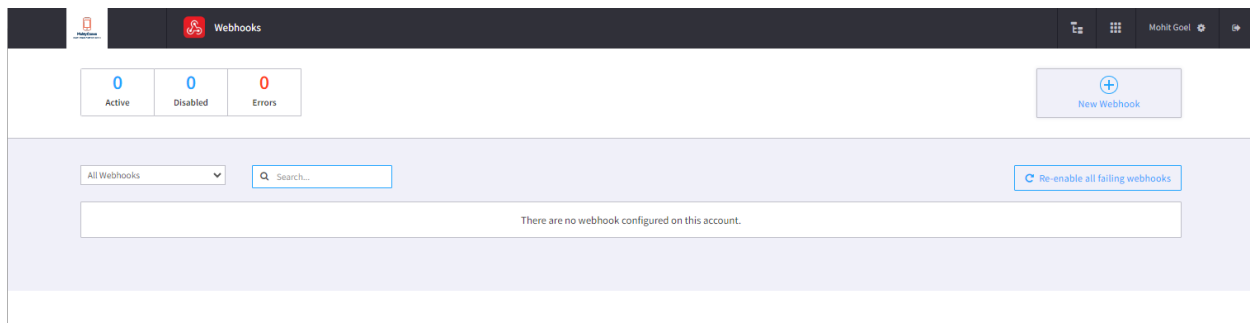
Webhook Integrations

Webhooks allow Multycomm to send HTTP requests to a third-party web server, alerting that server of events occurring within Multycomm. Typically, events would be fired for new calls, when a call is answered, and when a call is finished or when a configuration change has occurred.

Webhooks can be used to integrate with third party CRMs in order to auto populate customer information on incoming call or create popup alerts for Agents.

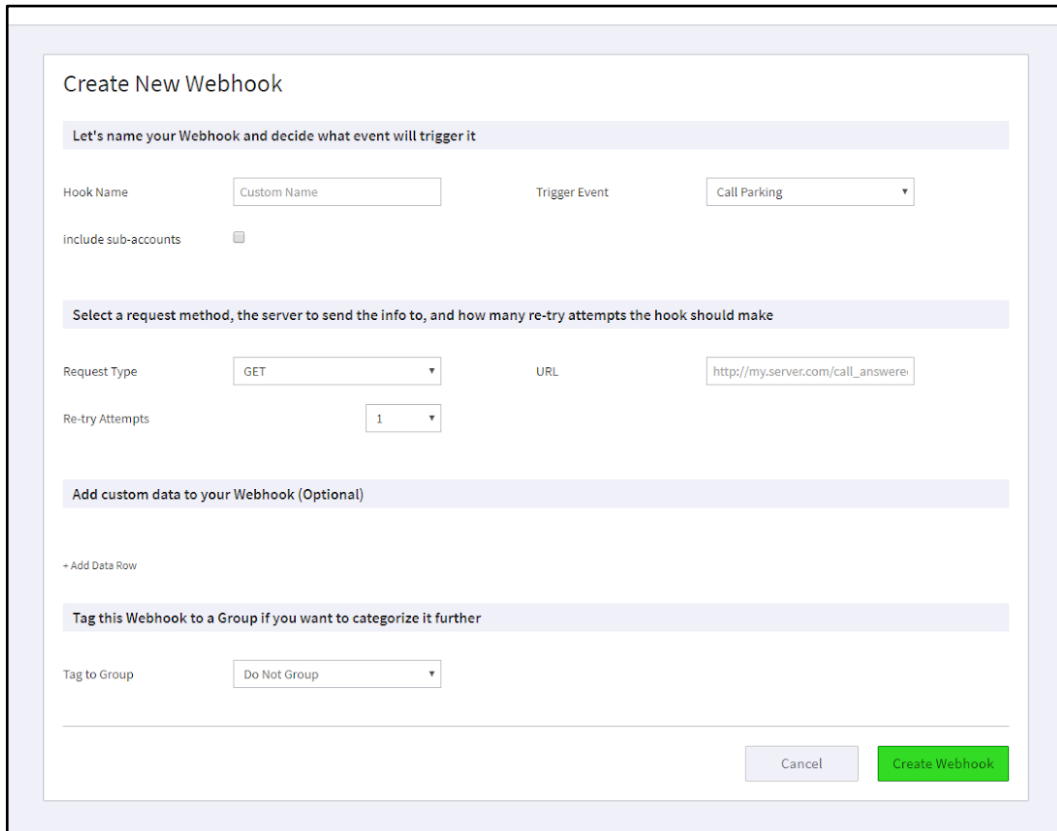
Open the Multycomm GUI

- Select Webhook Manager application
- Click on New Webhook in to right corner



Webhooks will auto-disable failing hooks (if Multycomm can't reach your server, or you take too long to respond with 200 OK, for instance). Especially if you're a reseller with webhooks in your client accounts, it can be tedious to have to iterate through all your accounts and re-enable each hook. Fortunately, you can perform this bulk-enable action against an account or an account and its descendants.

Create a new web hook



The screenshot shows a 'Create New Webhook' form with the following sections and fields:

- Let's name your Webhook and decide what event will trigger it**
 - Hook Name:
 - Trigger Event:
 - include sub-accounts: ☐
- Select a request method, the server to send the info to, and how many re-try attempts the hook should make**
 - Request Type:
 - URL:
 - Re-try Attempts:
- Add custom data to your Webhook (Optional)**
 - + Add Data Row
- Tag this Webhook to a Group if you want to categorize it further**
 - Tag to Group:

At the bottom right are two buttons: 'Cancel' and 'Create Webhook'.

Webhooks will track how many times a given hook has failed to fire to the configured URI. This expiry determines how long webhooks holds onto the failure before considering it irrelevant. The number of failures over a given time period will determine if a hook is auto-disabled.

Once you've set the expiry and threshold, webhooks will check every expiry milliseconds for hook failure counts. If any hooks have a count over threshold, they will be auto-disabled.

For instance, using the defaults of expiry = 60000 (one minute) and threshold = 6, if a hook fails to fire to the configured URI 6 times or more in the last minute, it will be auto-disabled until an API request to set "enabled" to true is performed.

Hook Payload

Here's what you can expect to receive when a webhook fires to your server:

Base Payload

Following fields will be present in every webhook payload.

hook_event:	The type of hook being fired
Call_direction:	"inbound" or "outbound", relative to Multycomm
timestamp:	Gregorian timestamp of the event
account_id:	ID of the account generating the event
request:	SIP Request
to:	SIP To
from:	SIP From
call_id:	SIP Call ID
other_leg_call_id:	If bridged, the Call ID of the other leg
caller_id_name:	Caller ID Name
caller_id_number:	Caller ID Number
callee_id_name:	Callee Name
callee_id_number:	Callee Number

Hook Specific

channel_create

Fired when an inbound call or outbound call is initiated, but before the call is bridged to the other party.

hook_event: channel_create

channel_answer

Fired when an inbound or outbound call is answered.

hook_event: channel_answer

channel_destroy

Fired when a call is hungup.

hook_event: channel_destroy
hangup_cause: SIP Hangup Cause (NORMAL_CLEARING, ORIGINATOR_CANCEL, etc)
hangup_code: SIP Hangup Code (404, 503, etc)

doc

Fired when configuration change occurs.

hook_event: doc
action: doc_created, doc_updated, doc_deleted

—

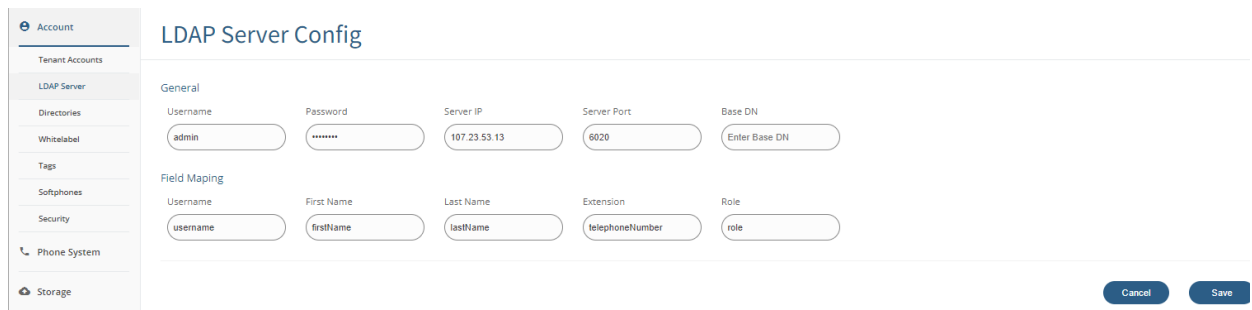
type: user, vmbox, callflow, account, device, faxbox, media

LDAP integration

In this section we will outline all the steps necessary to integrate your LDAP server and how to synchronize LDAP users as Multycomm CloudPBX extensions.

Configuration

1. Log into the Portal with the admin user of the account you're looking to add endpoints
 - a. **https://<domain>/portal**
 - b. User: your email
 - c. Pass: <password>
 - d. Account: Name of account without "space" or "-"
2. Select **Configuration** on the top navigation bar

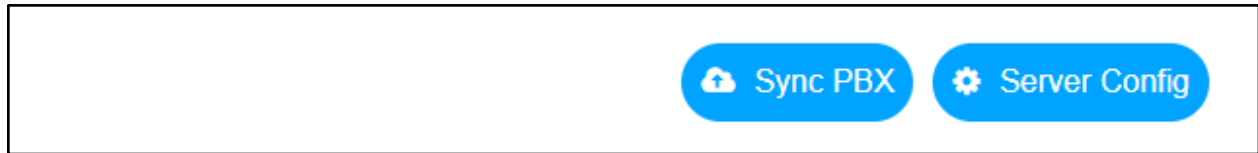


The screenshot shows the 'LDAP Server Config' form. On the left is a sidebar with a menu: Account (selected), Tenant Accounts, LDAP Server, Directories, Whitelabel, Tags, Softphones, Security, Phone System, and Storage. The main form area is titled 'LDAP Server Config' and contains two sections: 'General' and 'Field Mapping'. The 'General' section has five input fields: Username (admin), Password (masked with dots), Server IP (197.23.53.13), Server Port (6020), and Base DN (Enter Base DN). The 'Field Mapping' section has five input fields: Username (username), First Name (firstName), Last Name (lastName), Extension (telephoneNumber), and Role (role). At the bottom right of the form are 'Cancel' and 'Save' buttons.

3. Select LDAP on the *left* navigation bar
4. Click on LDAP server configuration ... button
 - a. Fill in LDAP server configuration section
 - i. Username: LDAP user authentication
 - ii. Password: LDAP password authentication
 - iii. Server IP: LDAP server IP address
 - iv. Port: LDAP server port
 - v. Base DN: LDAP distinguished name to target complete directory
 - b. Fill in Field Mapping section
 - i. Match desired LDAP fields to Multycomm CloudPBX user fields
5. Click on Save button
6. Configuration window will close and system will try to connect to LDAP server
7. If portal successfully connects to LDAP server and directory can be retrieved you will see the entire list otherwise you will see an error message
8. Now you may synchronize directory to Multycomm CloudPBX users
9. Refresh list button will request LDAP directory every time is clicked

Extensions synchronization

1. Once portal can connect to LDAP server and get directory Sync PBX Users button will be enabled



2. Click on Sync PBX Users button to synchronize LDAP users as PBX users
3. Users that do not exist on PBX will be created
4. If a user already exists it will be updated
5. If a PBX user was created from LDAP synchronization and he does not exist anymore on LDAP directory it will be deleted from PBX
6. PBX users created from sources other than LDAP will not be synchronized in any way
7. PBX valid roles are “**user**” or “**admin**”.
Setting user role anything other than admin, sets the role of the user to "user"

LDAP authentication on portal login

Once LDAP server configuration is set portal will authenticate users on login using LDAP server authentication.

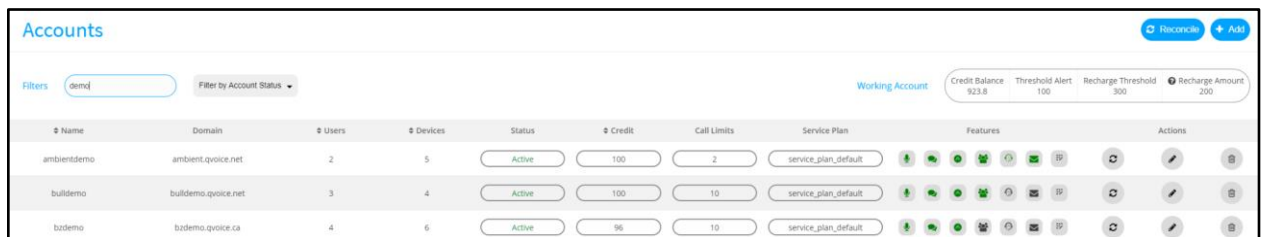
If LDAP authentication fails then portal internal authentication will be used.

Collaboration configuration

In this section we will outline all the steps necessary to enable collaboration feature for an account.

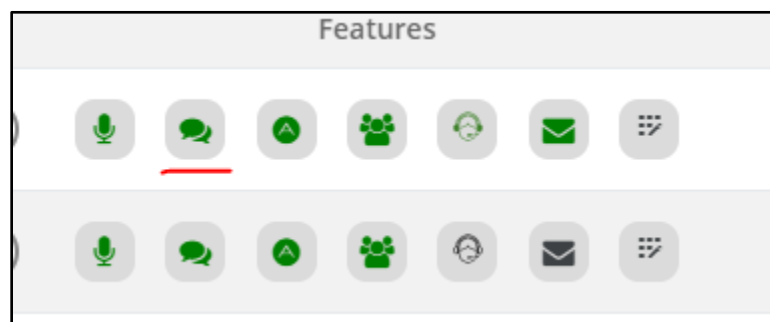
Configuration

1. Log into the Portal with the admin user of the account you're looking to add endpoints
 - a. **https://<domain>/portal**
 - b. User: your email
 - c. Pass: <password>
 - d. Account: Name of account without "space" or "-"
2. Select **Configuration** on the top navigation bar

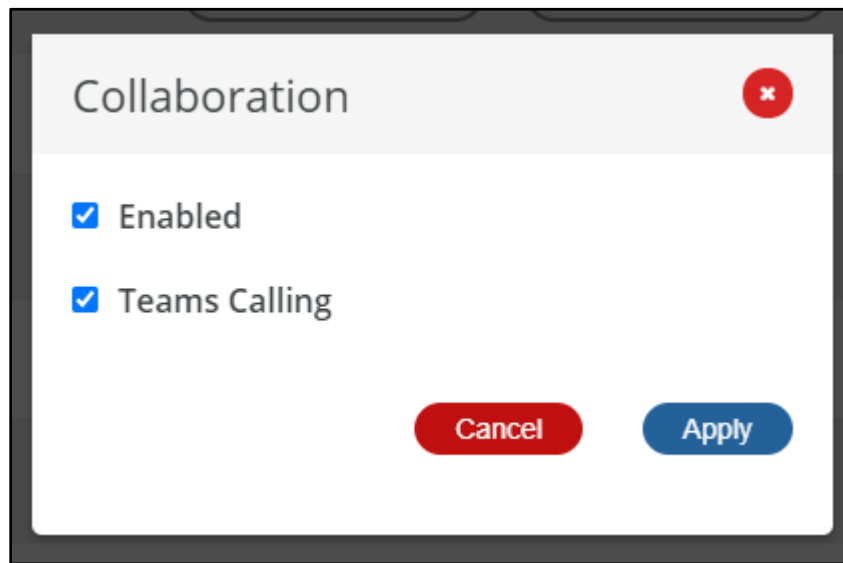


Name	Domain	Users	Devices	Status	Credit	Call Limits	Service Plan	Features	Actions
ambiendemo	ambient.voice.net	2	5	Active	100	2	service_plan_default		
bulldemo	bulldemo.voice.net	3	4	Active	100	10	service_plan_default		
budemo	budemo.voice.ca	4	6	Active	96	10	service_plan_default		

3. Select Accounts on the *left* navigation bar
4. Click on the pencil icon of the account you want to enable collaboration



5. Enable Collaboration



6. Save changes
7. Now Collaboration is enabled for account and all its users

Appendix

Troubleshooting

Desktop UCP Soft Client

Desktop softphones use the SIP/WebRTC technology to communicate to the Cloud UC platform. The Desktop softphone can be used as a Web browser application in Chrome or as a standalone Desktop Application.

Error: Cause Code	Destination	Type	Description
net::ERR_INCOMPLETE_CHUNKED_ENCODING	Outbound	Critical	Antivirus is blocking the Web or Desktop Softphone from establishing a two way websocket connection. Solution is to whitelist the Browser or Desktop application.

SIP Cause Codes

Cause Code	Destination	Type	Description
INCOMPATIBLE_DESTINATION	Outbound	Critical	Codec mismatch on Outbound Call Device configured for TLS/SRTP, client UDP
SERVICE_NOT_IMPLEMENTED	Inbound	Critical	Codec mismatch on Inbound Call Device configured for TLS/SRTP, client UDP
NORMAL_UNSPECIFIED	Inbound	Critical	The inbound number dialed is not configured or not bound to a user or a service. The number could be in "Spare numbers". System is trying to dial this number back out to PSTN, but user is unauthorized.

Hangup Codes

ITU-T Q.850 Code	SIP Equiv.	Enumeration	Cause	Description
0		UNSPECIFIED	Unspecified. No other cause codes applicable.	This is usually given by the router when none of the other codes apply. This cause usually occurs in the same type of situations as cause 1, cause 88, and cause 100.

1	404	<i>UNALLOCATED_NUMBER</i>	Unallocated (unassigned) number [Q.850 value 1]	This cause indicates that the called party cannot be reached because, although the called party number is in a valid format, it is not currently allocated (assigned).
2	404	<i>NO_ROUTE_TRANSIT_NET</i>	No route to specified transit network (national use) [Q.850]	This cause indicates that the equipment sending this cause has received a request to route the call through a particular transit network, which it does not recognize. The equipment sending this cause does not recognize the transit network either because the transit network does not exist or because that particular transit network, while it does exist, does not serve the equipment which is sending this cause.
3	404	<i>NO_ROUTE_DESTINATION</i>	No route to destination [Q.850]	This cause indicates that the called party cannot be reached because the network through which the call has been routed does not serve the destination desired. This cause is supported on a network dependent basis.
6		<i>CHANNEL_UNACCEPTABLE</i>	channel unacceptable [Q.850]	This cause indicates that the channel most recently identified is not acceptable to the sending entity for use in this call.
7		<i>CALL_AWARDED_DELIVERED</i>	call awarded, being delivered in an established channel [Q.850]	This cause indicates that the user has been awarded the incoming call, and that the incoming call is being connected to a channel already established to that user for similar calls (e.g. packet-mode x.25 virtual calls).
16		<i>NORMAL_CLEARING</i>	normal call clearing [Q.850]	This cause indicates that the call is being cleared because one of the users involved in the call has requested that the call be cleared. Under normal situations, the source of this cause is not the network.
17	486	<i>USER_BUSY</i>	user busy [Q.850]	This cause is used to indicate that the called party is unable to accept another call because the user busy condition has been encountered. This cause value may be generated by the called user or by the network. In the case of user determined user busy it is noted that the user equipment is compatible with the call.
18	408	<i>NO_USER_RESPONSE</i>	no user responding [Q.850]	This cause is used when a called party does not respond to a call establishment message with either an alerting or connect indication within the prescribed period of time allocated.

19	480	<i>NO_ANSWER</i>	no answer from user (user alerted) [Q.850]	This cause is used when the called party has been alerted but does not respond with a connect indication within a prescribed period of time. Note - This cause is not necessarily generated by Q.931 procedures but may be generated by internal network timers.
20	480	<i>SUBSCRIBER_ABSENT</i>	subscriber absent [Q.850]	This cause value is used when a mobile station has logged off, radio contact is not obtained with a mobile station or if a personal telecommunication user is temporarily not addressable at any user-network interface. Sofia SIP will normally raise USER_NOT_REGISTERED in such situations.
21	603	<i>CALL_REJECTED</i>	call rejected [Q.850]	This cause indicates that the equipment sending this cause does not wish to accept this call, although it could have accepted the call because the equipment sending this cause is neither busy nor incompatible. The network may also generate this cause, indicating that the call was cleared due to a supplementary service constraint. The diagnostic field may contain additional information about the supplementary service and reason for rejection.
22	410	<i>NUMBER_CHANGED</i>	number changed [Q.850]	This cause is returned to a calling party when the called party number indicated by the calling party is no longer assigned. The new called party number may optionally be included in the diagnostic field. If a network does not support this cause, cause no: 1, unallocated (unassigned) number shall be used.
23	410	<i>REDIRECTION_TO_NEW_DESTINATION</i>		This cause is used by a general ISUP protocol mechanism that can be invoked by an exchange that decides that the call should be set-up to a different called number. Such an exchange can invoke a redirection mechanism, by use of this cause value, to request a preceding exchange involved in the call to route the call to the new number.
25	483	<i>EXCHANGE_ROUTING_ERROR</i>		This cause indicates that the destination indicated by the user cannot be reached, because an intermediate exchange has released the call due to reaching a limit in executing the hop counter procedure. This cause is generated by an intermediate node, which when decrementing the hop counter value, gives the result 0.
27	502	<i>DESTINATION_OUT_OF_ORDER</i>	destination out of order [Q.850]	This cause indicates that the destination indicated by the user cannot be reached because the interface to the destination is not functioning correctly. The term "not functioning correctly" indicates that a signal message was unable to be delivered to the remote party; e.g. a physical layer or data link layer failure at the remote party, or user equipment off-line.

28	484	<i>INVALID_NUMBER_FORMAT</i>	invalid number format (address incomplete) [Q.850]	This cause indicates that the called party cannot be reached because the called party number is not in a valid format or is not complete.
29	501	<i>FACILITY_REJECTED</i>	facilities rejected [Q.850]	This cause is returned when a supplementary service requested by the user cannot be provided by the network.
30		<i>RESPONSE_TO_STATUS_ENQUIRY</i>	response to STATUS INQUIRY [Q.850]	This cause is included in the STATUS message when the reason for generating the STATUS message was the prior receipt of a STATUS INQUIRY.
31	480	<i>NORMAL_UNSPECIFIED</i>	normal, unspecified [Q.850]	This cause is used to report a normal event only when no other cause in the normal class applies.
34	503	<i>NORMAL_CIRCUIT_CONGESTION</i>	no circuit/channel available [Q.850]	This cause indicates that there is no appropriate circuit/channel presently available to handle the call.
38	502	<i>NETWORK_OUT_OF_ORDER</i>	network out of order [Q.850]	This cause indicates that the network is not functioning correctly and that the condition is likely to last a relatively long period of time e.g. immediately re-attempting the call is not likely to be successful.
41	503	<i>NORMAL_TEMPORARY_FAILURE</i>	temporary failure [Q.850]	This cause indicates that the network is not functioning correctly and that the condition is not likely to last a long period of time; e.g. the user may wish to try another call attempt almost immediately.
42	503	<i>SWITCH_CONGESTION</i>	switching equipment congestion [Q.850]	This cause indicates that the switching equipment generating this cause is experiencing a period of high traffic.
43		<i>ACCESS_INFO_DISCARDED</i>	access information discarded [Q.850]	This cause indicates that the network could not deliver access information to the remote user as requested, i.e. user-to-user information, low layer compatibility, high layer compatibility or sub-address as indicated in the diagnostic. It is noted that the particular type of access information discarded is optionally included in the diagnostic.

44	503	<i>REQUESTED_CHAN_UNAVAIL</i>	requested circuit/channel not available [Q.850]	This cause is returned when the other side of the interface cannot provide the circuit or channel indicated by the requesting entity.
45		<i>PRE_EMPTED</i>		
47			resource unavailable, unspecified [Q.850]	This cause is used to report a resource unavailable event only when no other cause in the resource unavailable class applies.
50		<i>FACILITY_NOT_SUBSCRIBED</i>	requested facility not subscribed [Q.850]	This cause indicates that the user has requested a supplementary service, which is available, but the user is not authorized to use.
52	403	<i>OUTGOING_CALL_BARRED</i>	outgoing calls barred	This cause indicates that although the calling party is a member of the CUG for the outgoing CUG call, outgoing calls are not allowed for this member of the CUG.
54	403	<i>INCOMING_CALL_BARRED</i>	incoming calls barred	This cause indicates that although the called party is a member of the CUG for the incoming CUG call, incoming calls are not allowed to this member of the CUG.
57	403	<i>BEARERCAPABILITY_NOTAUTH</i>	bearer capability not authorized [Q.850]	This cause indicates that the user has requested a bearer capability that is implemented by the equipment which generated this cause but the user is not authorized to use.
58	503	<i>BEARERCAPABILITY_NOTAVAIL</i>	bearer capability not presently available [Q.850]	This cause indicates that the user has requested a bearer capability which is implemented by the equipment which generated this cause but which is not available at this time.
63		<i>SERVICE_UNAVAILABLE</i>	service or option not available, unspecified [Q.850]	This cause is used to report a service or option not available event only when no other cause in the service or option not available class applies.
65	488	<i>BEARERCAPABILITY_NOTIMPL</i>	bearer capability not implemented [Q.850]	This cause indicates that the equipment sending this cause does not support the bearer capability requested.

66		<i>CHAN_NOT_IMPLEMENTED</i>	channel type not implemented [Q.850]	This cause indicates that the equipment sending this cause does not support the channel type requested
69	501	<i>FACILITY_NOT_IMPLEMENTED</i>	requested facility not implemented [Q.850]	This cause indicates that the equipment sending this cause does not support the requested supplementary services.
79	501	<i>SERVICE_NOT_IMPLEMENTED</i>	service or option not implemented, unspecified [Q.850]	This cause is used to report a service or option not implemented event only when no other cause in the service or option not implemented class applies.
81		<i>INVALID_CALL_REFERENCE</i>	invalid call reference value [Q.850]	This cause indicates that the equipment sending this cause has received a message with a call reference which is not currently in use on the user-network interface.
88	488	<i>INCOMPATIBLE_DESTINATION</i>	incompatible destination [Q.850]	This cause indicates that the equipment sending this cause has received a request to establish a call which has low layer compatibility, high layer compatibility or other compatibility attributes (e.g. data rate) which cannot be accommodated.
95		<i>INVALID_MSG_UNSPECIFIED</i>	invalid message, unspecified [Q.850]	This cause is used to report an invalid message event only when no other cause in the invalid message class applies.
96		<i>MANDATORY_IE_MISSING</i>	mandatory information element is missing [Q.850]	This cause indicates that the equipment sending this cause has received a message which is missing an information element which must be present in the message before that message can be processed.
97		<i>MESSAGE_TYPE_NONEXIST</i>	message type non-existent or not implemented [Q.850]	This cause indicates that the equipment sending this cause has received a message with a message type it does not recognize either because this is a message not defined or defined but not implemented by the equipment sending this cause.

98		<i>WRONG_MESSAGE</i>	message not compatible with call state or message type non-existent or not implemented. [Q.850]	This cause indicates that the equipment sending this cause has received a message such that the procedures do not indicate that this is a permissible message to receive while in the call state, or a STATUS message was received indicating an incompatible call state.
99		<i>IE_NONEXIST</i>	Information element / parameter non-existent or not implemented [Q.850]	This cause indicates that the equipment sending this cause has received a message which includes information element(s)/parameter(s) not recognized because the information element(s)/parameter name(s) are not defined or are defined but not implemented by the equipment sending the cause. This cause indicates that the information element(s)/parameter(s) were discarded. However, the information element is not required to be present in the message in order for the equipment sending the cause to process the message.
100		<i>INVALID_IE_CONTENTS</i>	Invalid information element contents [Q.850]	This cause indicates that the equipment sending this cause has received and information element which it has implemented; however, one or more fields in the I.E. are coded in such a way which has not been implemented by the equipment sending this cause.
101		<i>WRONG_CALL_STATE</i>	message not compatible with call state [Q.850]	This cause indicates that a message has been received which is incompatible with the call state.
102	504	<i>RECOVERY_ON_TIMER_EXPIRE</i>	recovery on timer expiry [Q.850]	This cause indicates that a procedure has been initiated by the expiration of a timer in association with error handling procedures. This is often associated with NAT problems. Ensure that "NAT Mapping Enable" is turned on in your ATA. If it is not NAT related it can sometimes be provider related, make sure to ensure another outbound provider does not solve the problem. FreeSWITCH also returns this when the remote party sends a 408 for call expired.

103		<i>MANDATORY_IE_LENGTH_ERROR</i>	parameter non-existent or not implemented - passed on (national use) [Q.850]	This cause indicates that the equipment sending this cause has received a message which includes parameters not recognized because the parameters are not defined or are defined but not implemented by the equipment sending this cause. The cause indicates that the parameter(s) were ignored. In addition, if the equipment sending this cause is an intermediate point, then this cause indicates that the parameter(s) were passed unchanged.
111		<i>PROTOCOL_ERROR</i>	protocol error, unspecified [Q.850]	This cause is used to report a protocol error event only when no other cause in the protocol error class applies.
127		<i>INTERWORKING</i>	Interworking, unspecified [Q.850]	This cause indicates that an interworking call (usually a call to SW56 service) has ended.
487	487	<i>ORIGINATOR_CANCEL</i>		
500		<i>CRASH</i>		
501		<i>SYSTEM_SHUTDOWN</i>		
502		<i>LOSE_RACE</i>		
503		<i>MANAGER_REQUEST</i>		This cause is used when you send an api command to make it hangup. For example uuid_kill <uuid>
600		<i>BLIND_TRANSFER</i>		
601		<i>ATTENDED_TRANSFER</i>		
602		<i>ALLOTTED_TIMEOUT</i>		This cause means that the server canceled the call because the destination channel took too long to answer.
603		<i>USER_CHALLENGE</i>		
604		<i>MEDIA_TIMEOUT</i>		

605		<i>PICKED_OFF</i>		This cause means the call was picked up by intercepting it from another extension (i.e. dialing **ext_number from another extension).
606		<i>USER_NOT_REGISTERED</i>		This means you tried to originate a call to a SIP user who forgot to register.
607		<i>PROGRESS_TIMEOUT</i>		See: progress_timeout
609		<i>GATEWAY_DOWN</i>		Gateway is down (not answering on OPTIONS or SUBSCRIBE)

SIP to Q.850 Code Table

These mappings are taken from [RFC 4497](#) section 8.4.4.

SIP Code	ITU-T Q.850 Code	Enumeration	Cause
400	41	<i>NORMAL_TEMPORARY_FAILURE</i>	Temporary failure [Q.850]
401	21	<i>CALL_REJECTED</i>	Call rejected [Q.850]
402	21	<i>CALL_REJECTED</i>	Call rejected [Q.850]
403	21	<i>CALL_REJECTED</i>	Call rejected [Q.850]
404	1	<i>NO_ROUTE_TRANSIT_NET</i>	Unallocated number [Q.850]